Hysteresis Indicies and Beta of IBW sites: Curry, Silverado, and Lake Marguerite

2026-03-26

Aims to plot hysteresis per analyte per storm per site

Metrics

- *Hysteresis index* is calculated as the median (+/- 95% CI) across 2% intervals of discharge.

  +HI = clockwise (source limitation or nearstream source)

  -HI = counterclockwise (transport limitation or distal source)

- *Beta* is the slope of the CQ relationship on the rising limb of the storm

  CIs on beta are bootstrapped 95% CI

NOTES

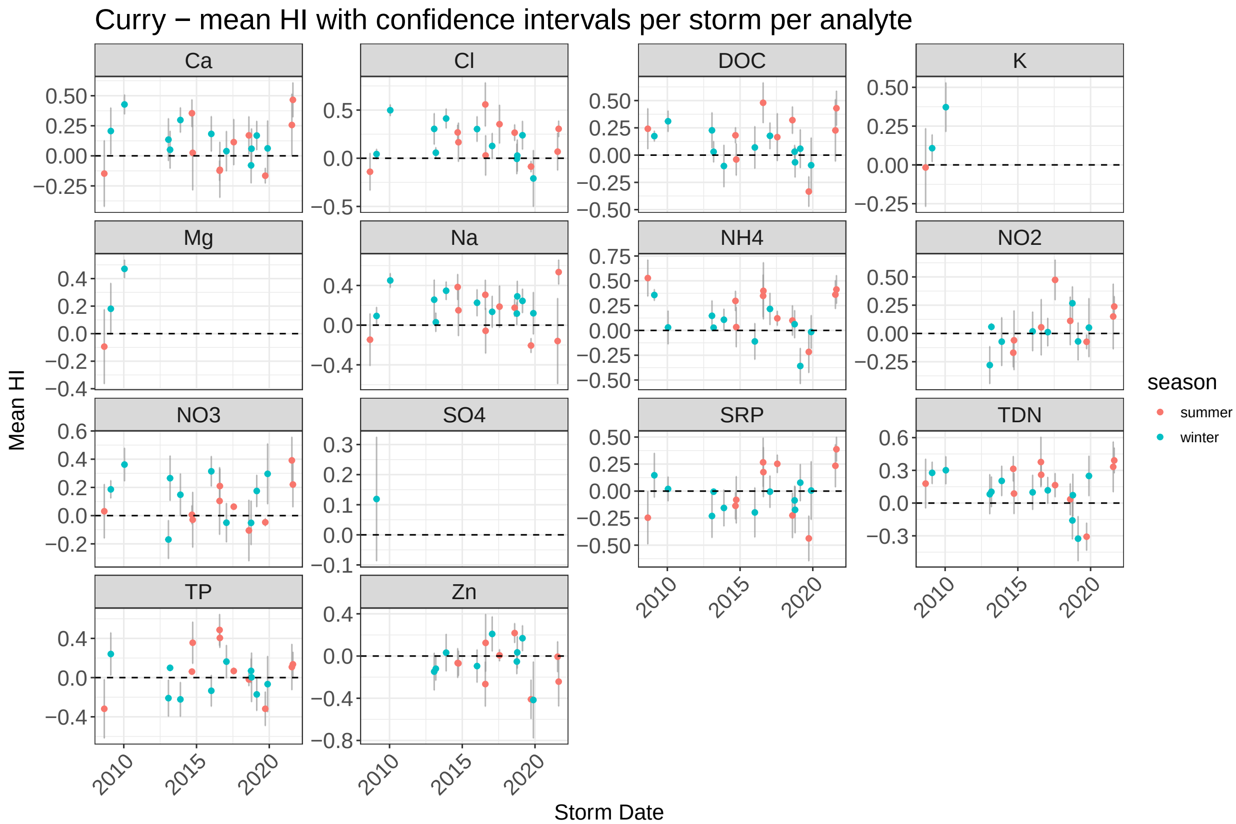
- Lake Marguerite does not have a lot of chems to work with. Do we know if there are more chems than the 624_runoff_chems.csv from the dropbox?
- Similarly, there is a gap of data in Silverado 2014-2018, was there less sampling in that time or is the data elsewhere?
- individual analyte outliers > 3sd above mean concentration per storm were removed for hysteresis and beta calculations, but remain for individual storm plotting at the bottom of the report
- Precipitation data was collected from https://azmet.arizona.edu/azmet/27.htm - precipitation gage at the Desert Ridge station

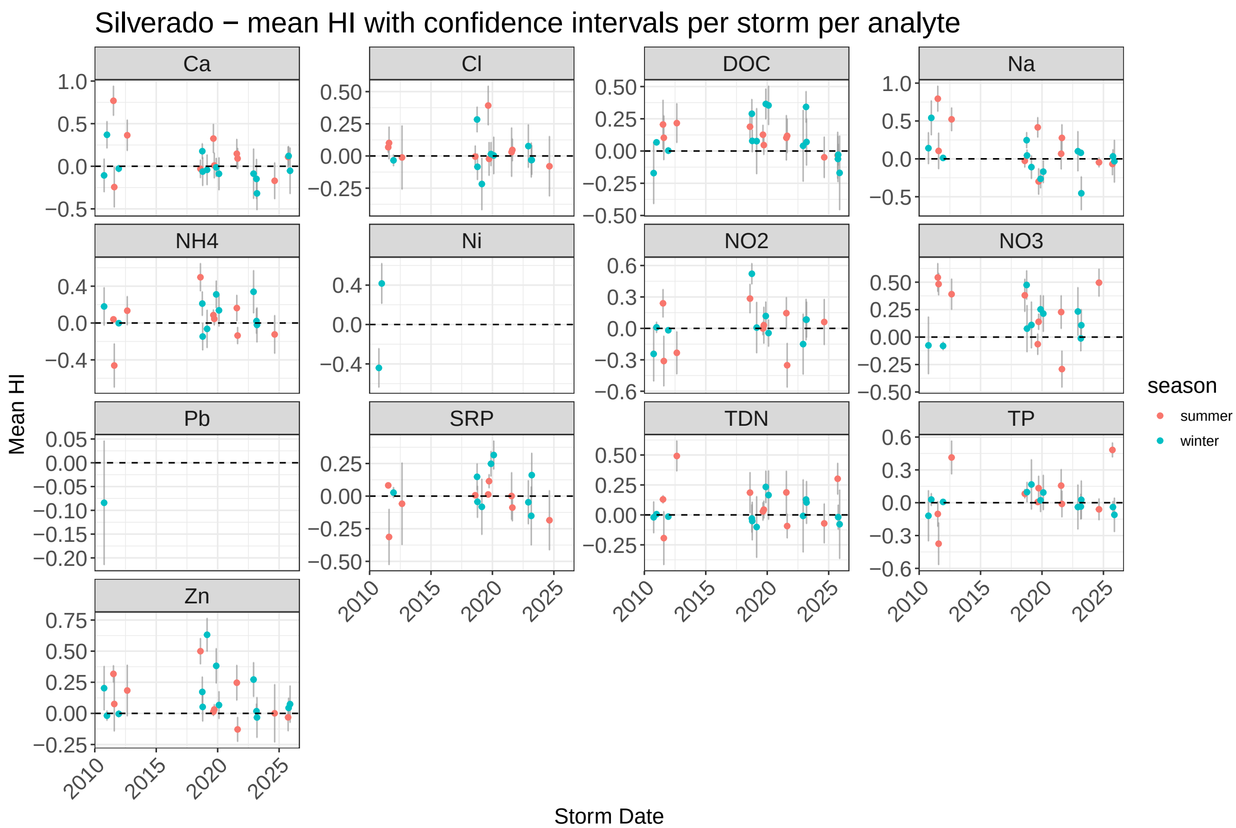
Data munging/cleaning

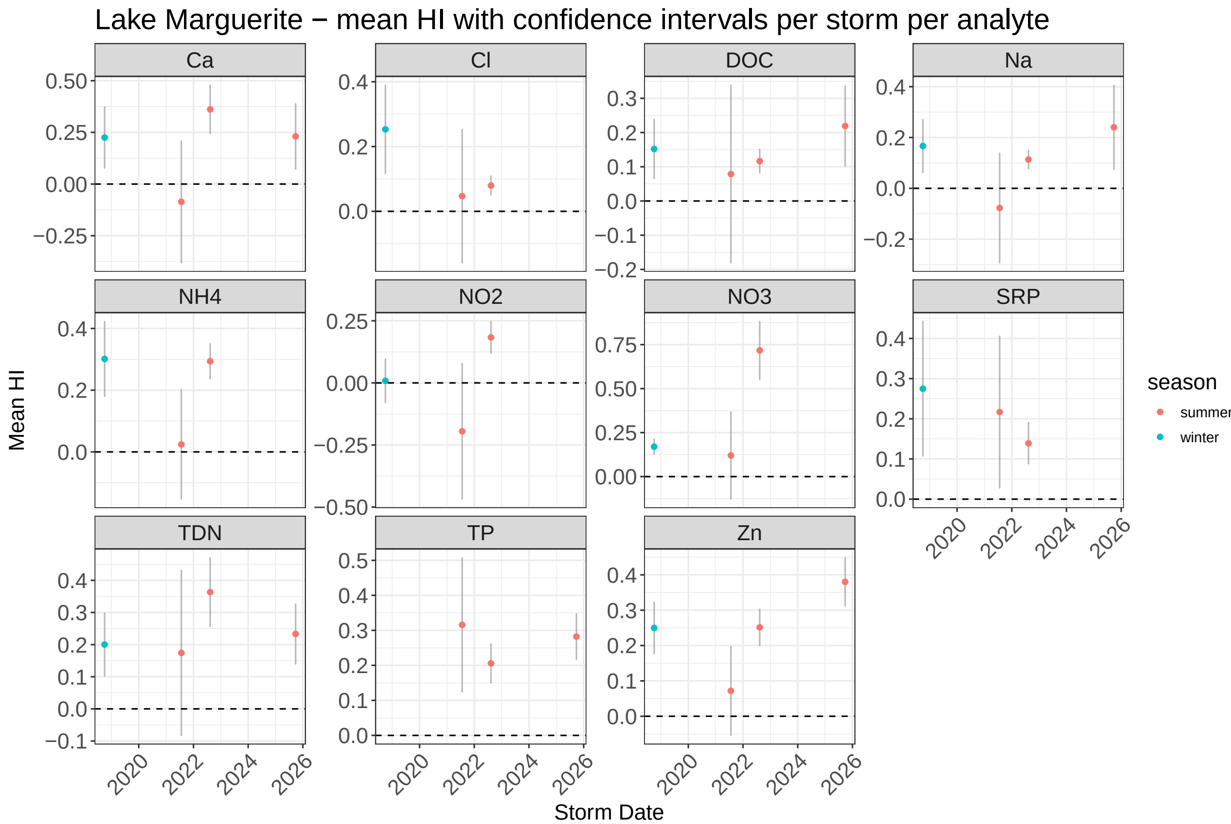
Workflow before this rmd, in order:

- q_storms.R: import discharge from Lake Margherite, Silverado, & Curry, delineate unique storms, and interpolate Q
- precipitation.R: import precip data and merge with Q to calculate pre-storm precip metrics
- merge_Q_chems.R: merge discharge data with chem data per site, produce brief cq plots per analyte per site
- hysteresis_calc_2.R: calculate hysteresis based on python function, removed a lot of storms with not enough data. also calculates beta

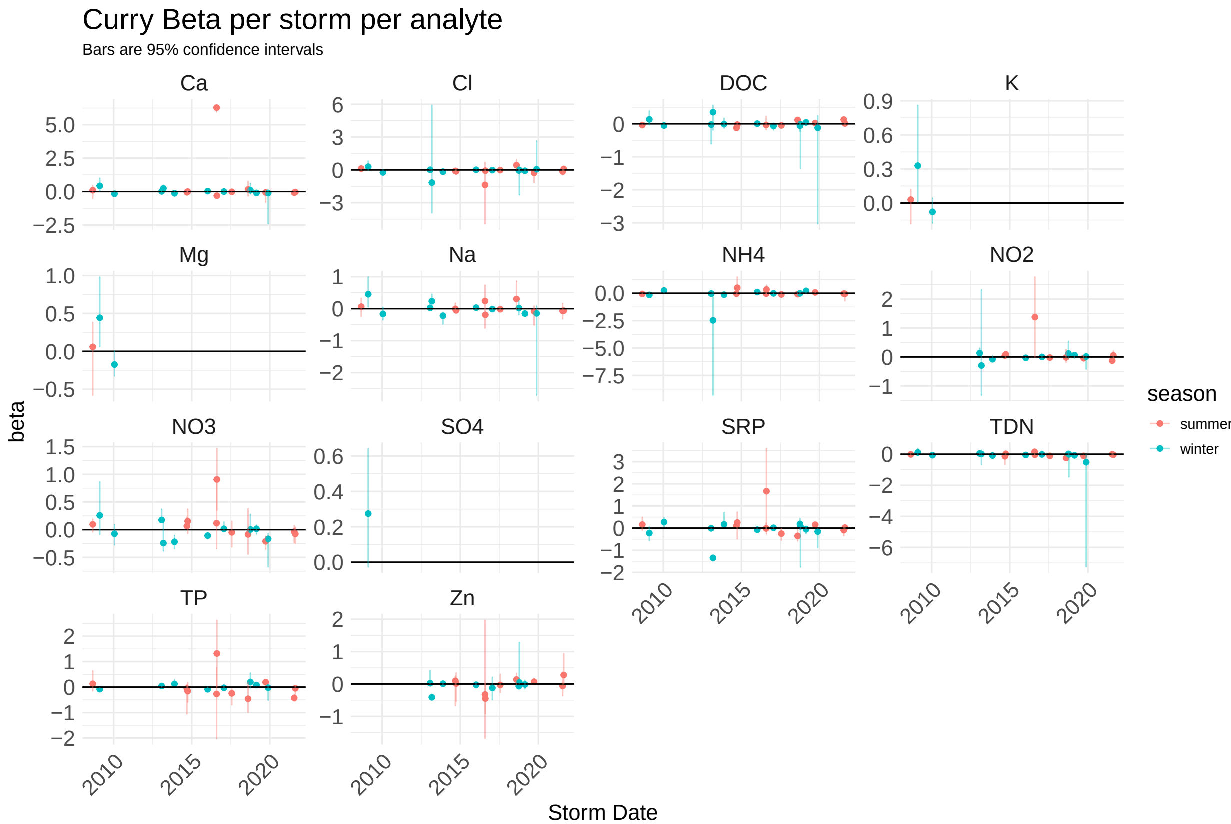
Plotting Hysteresis

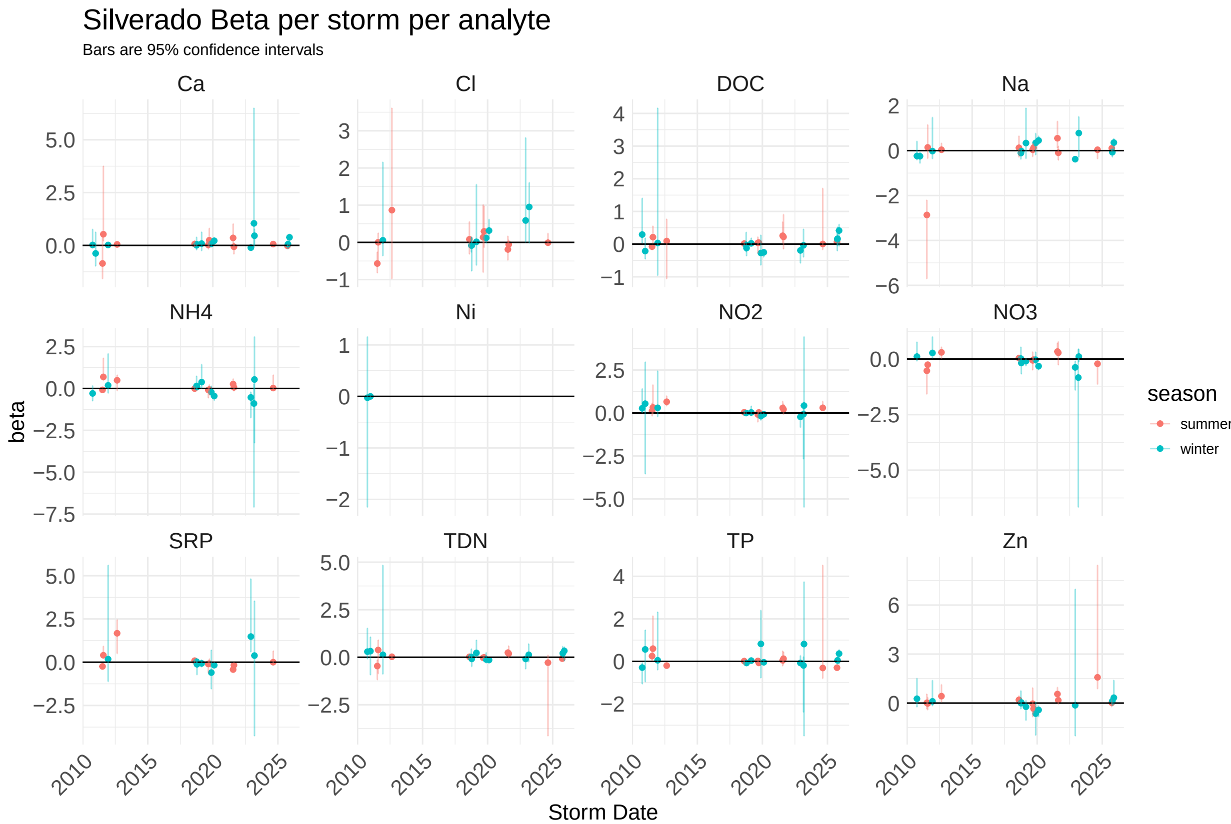


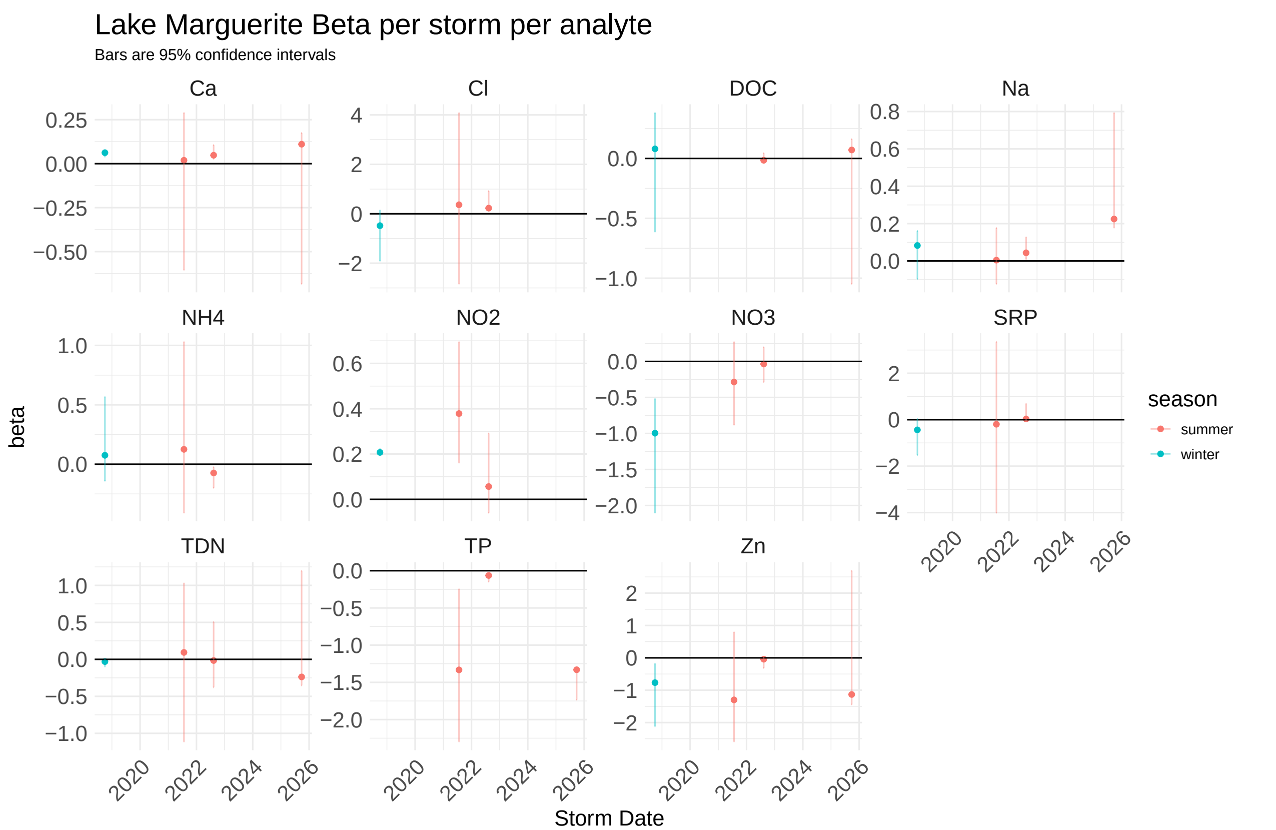



Lake Marguerite − mean HI with confidence intervals per storm per analyte
Ca
Cl
DOC
Na
NH4
NO2
NO3
SRP
TDN
TP
Zn
Mean HI
Storm Date
season
summer
winter

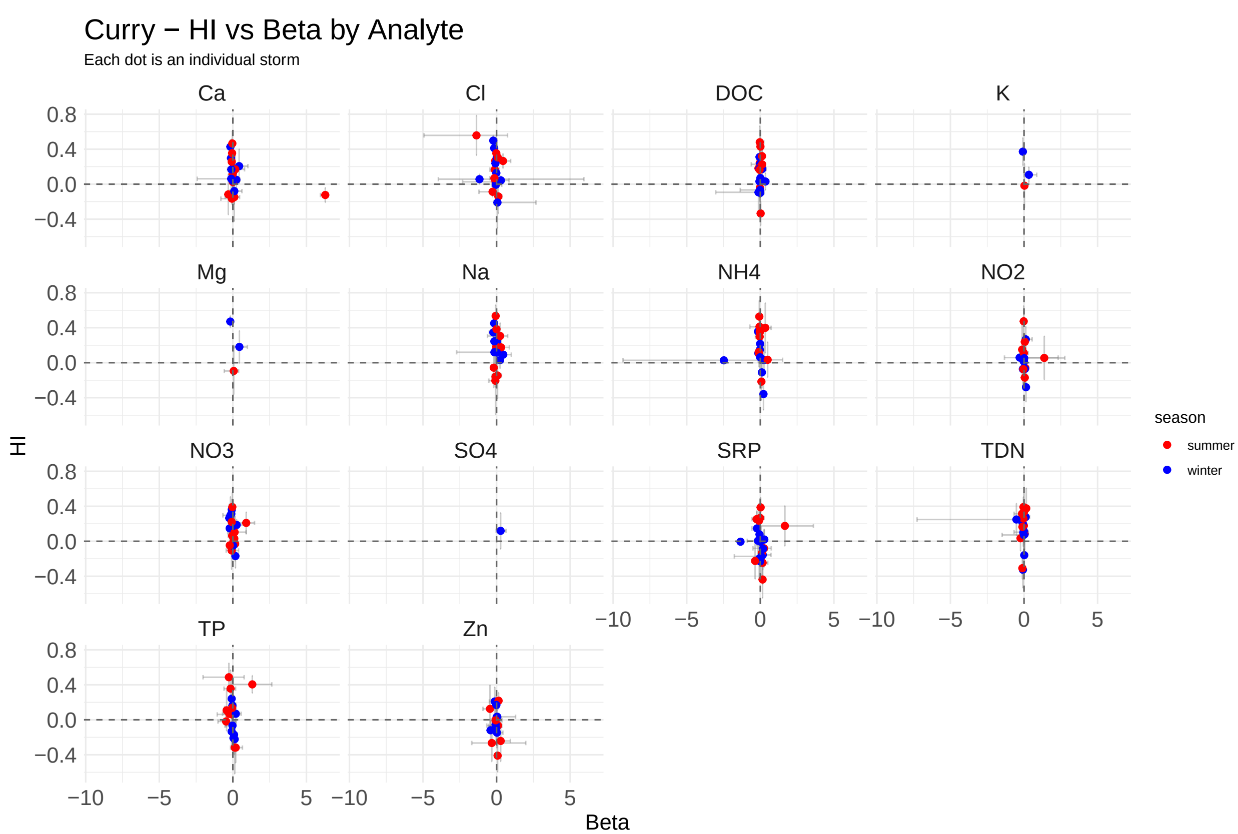
Plotting Beta







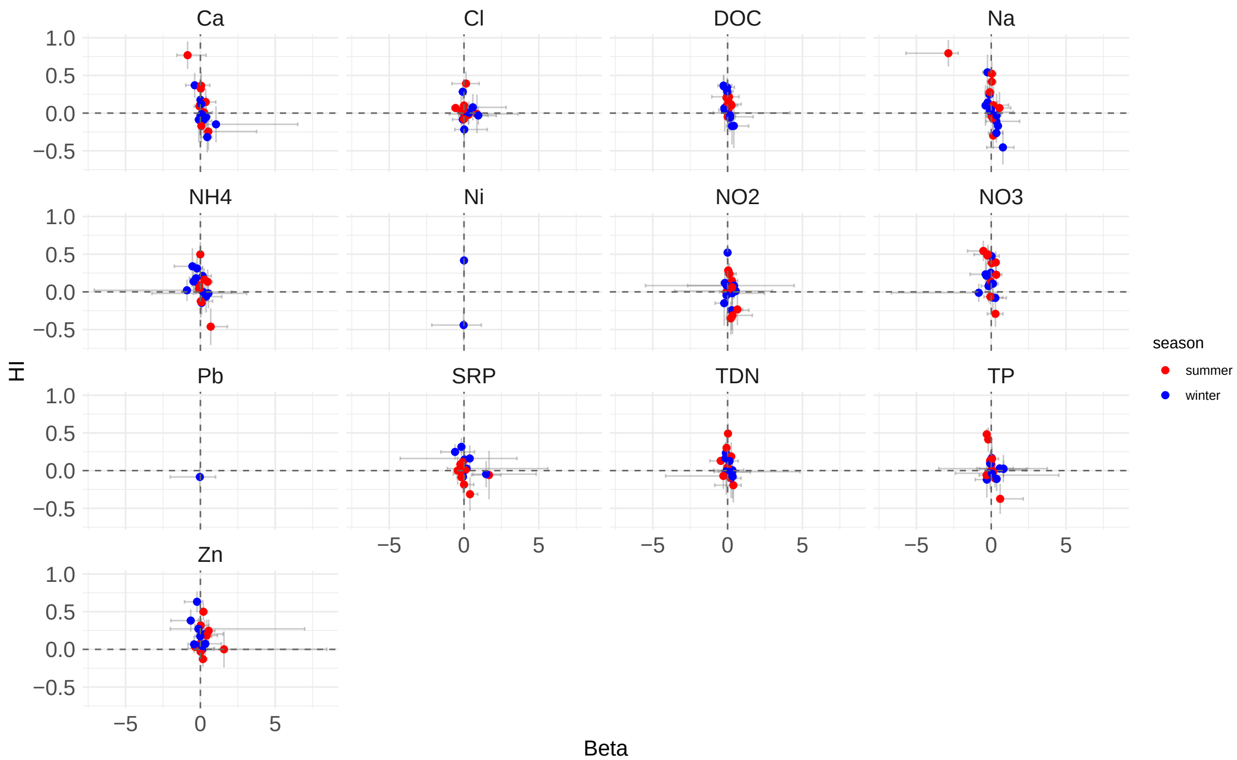
Lake Marguerite Beta per storm per analyte
Bars are 95% confidence intervals
Ca
Cl
DOC
Na
NH4
NO2
NO3
SRP
TDN
TP
Zn
beta
Storm Date
season
summer
winter



Curry − HI vs Beta by Analyte
Each dot is an individual storm
Ca
Cl
DOC
K
Mg
Na
NH4
NO2
NO3
SO4
SRP
TDN
TP
Zn
HI
Beta
season
summer
winter

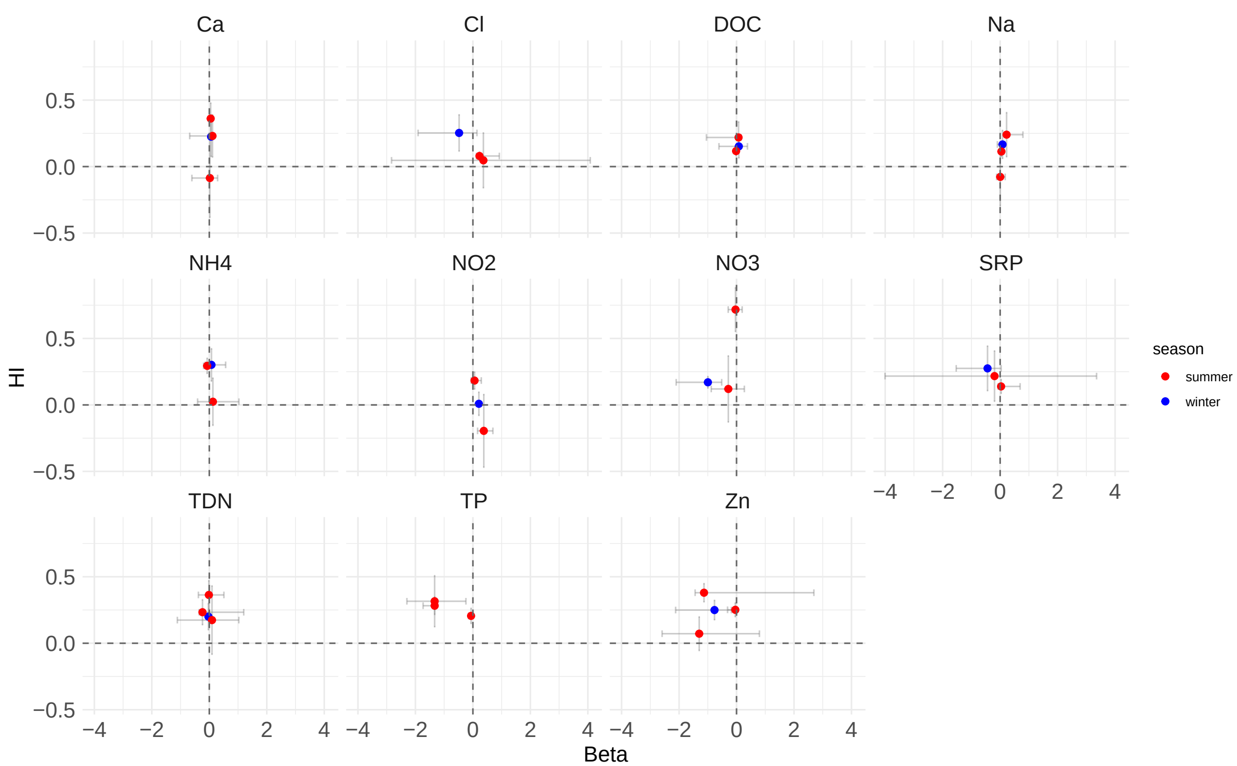
Silverado − HI vs Beta by Analyte

Each dot is an individual storm

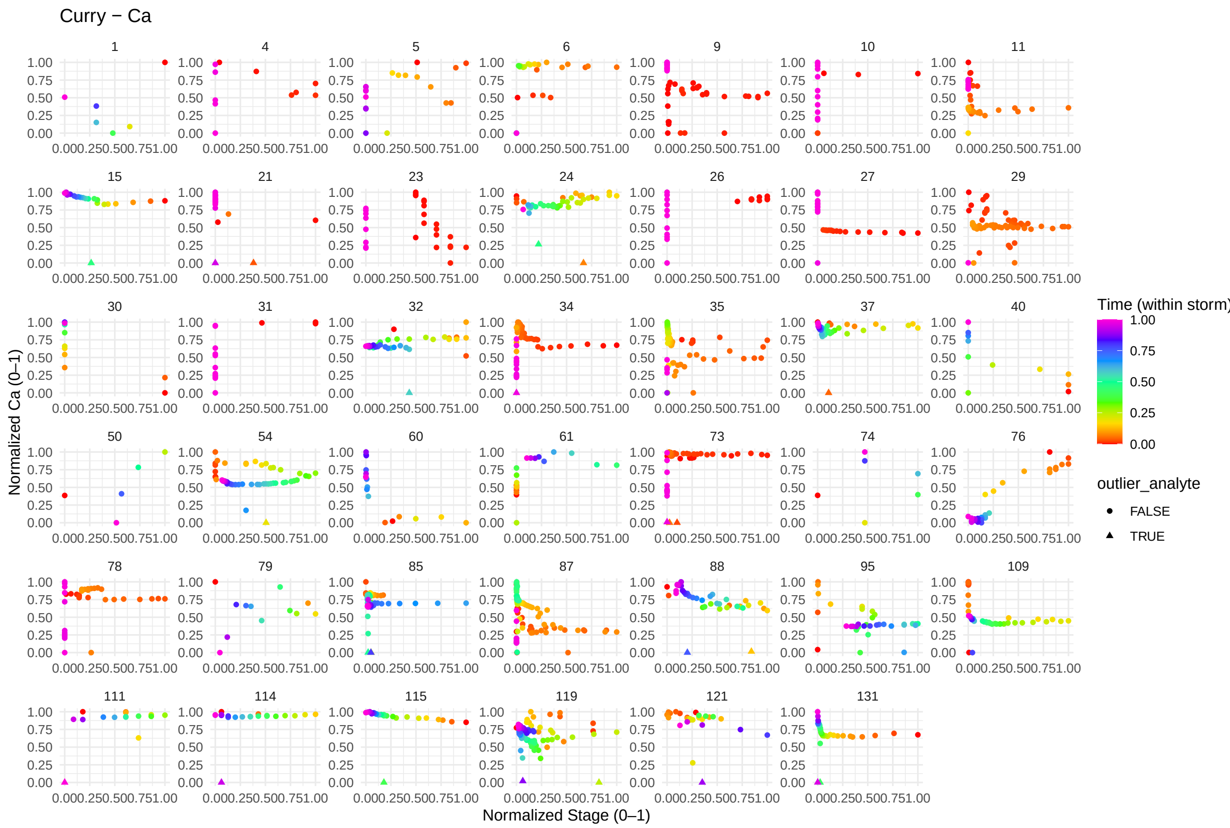


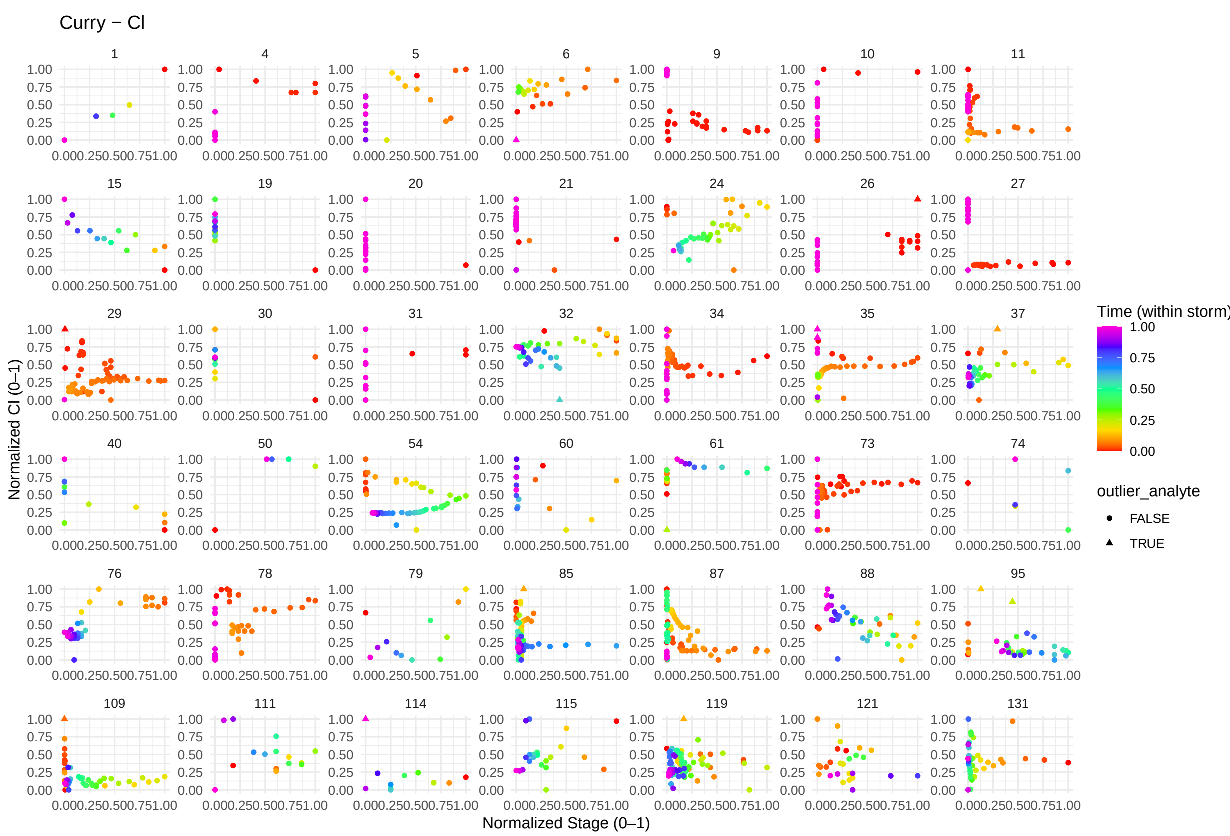
Lake Margarita − HI vs Beta by Analyte

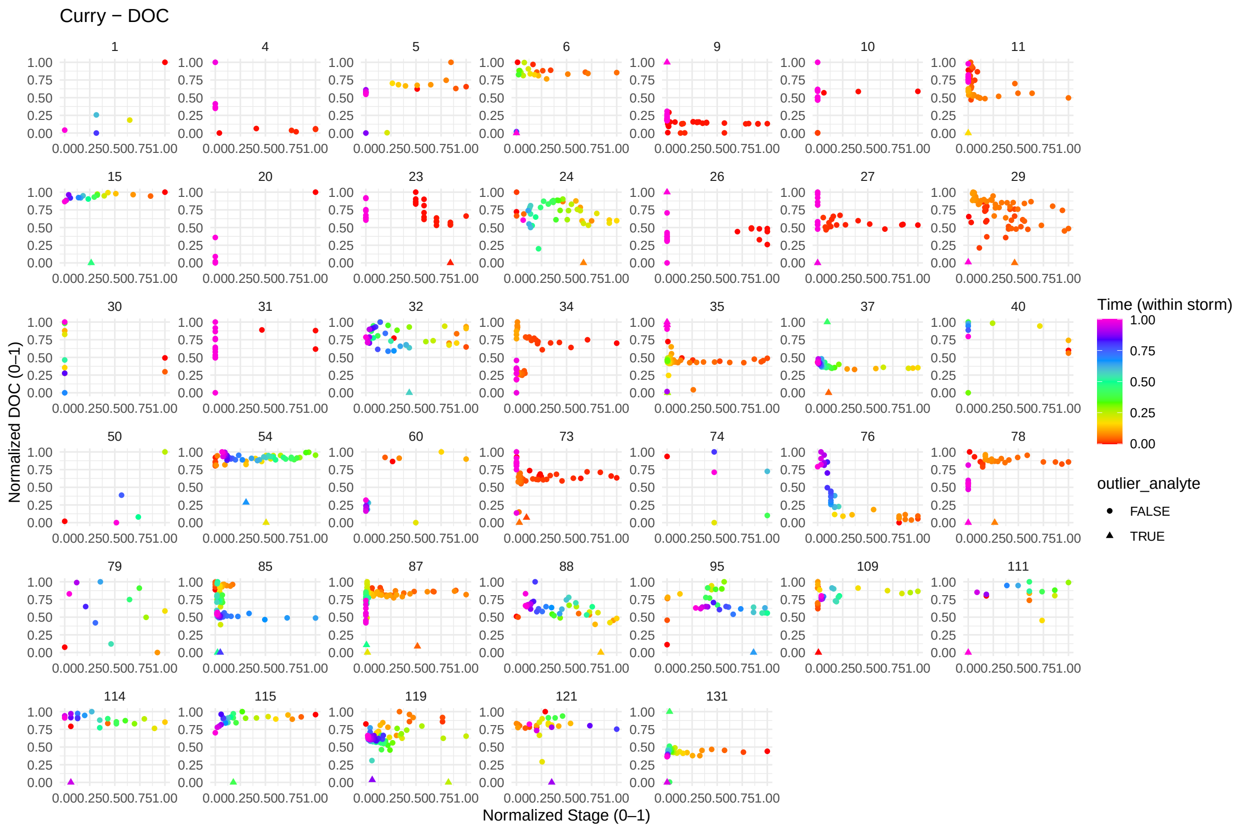
Each dot is an individual storm

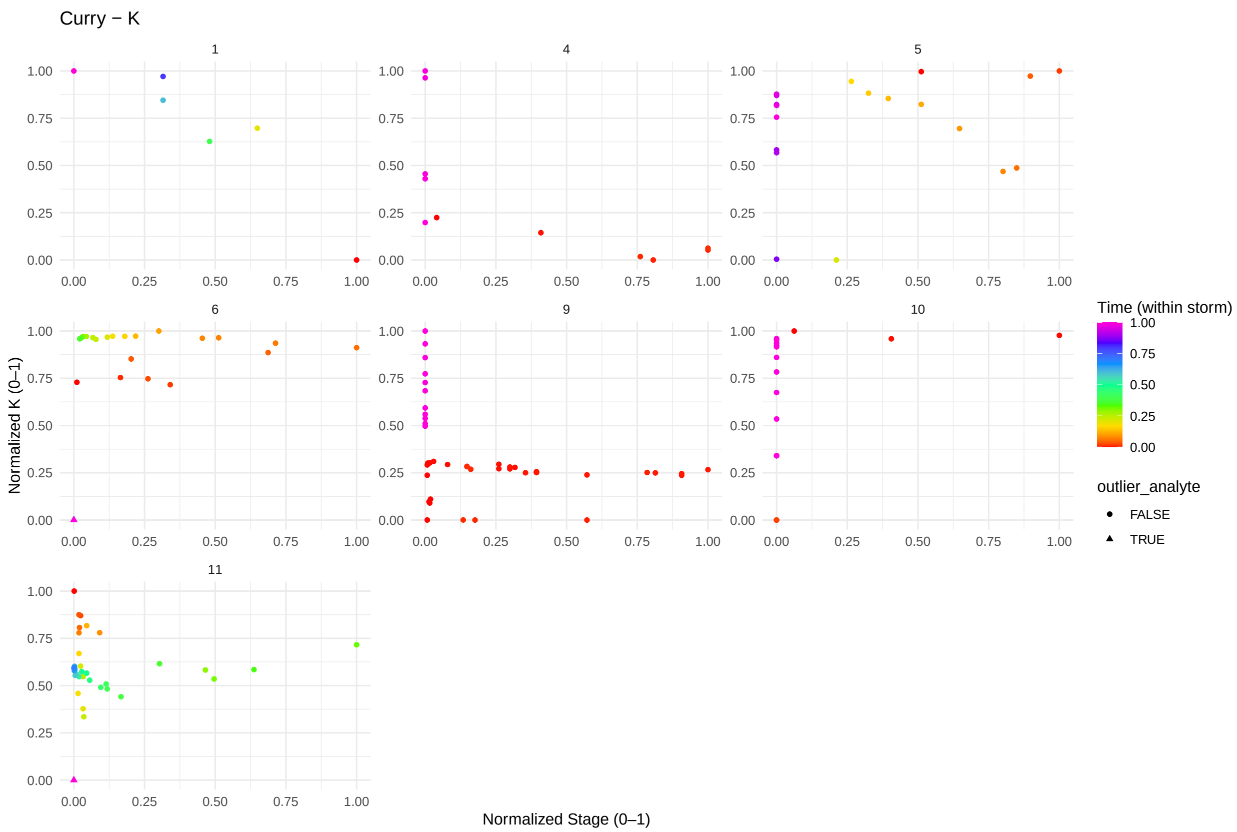


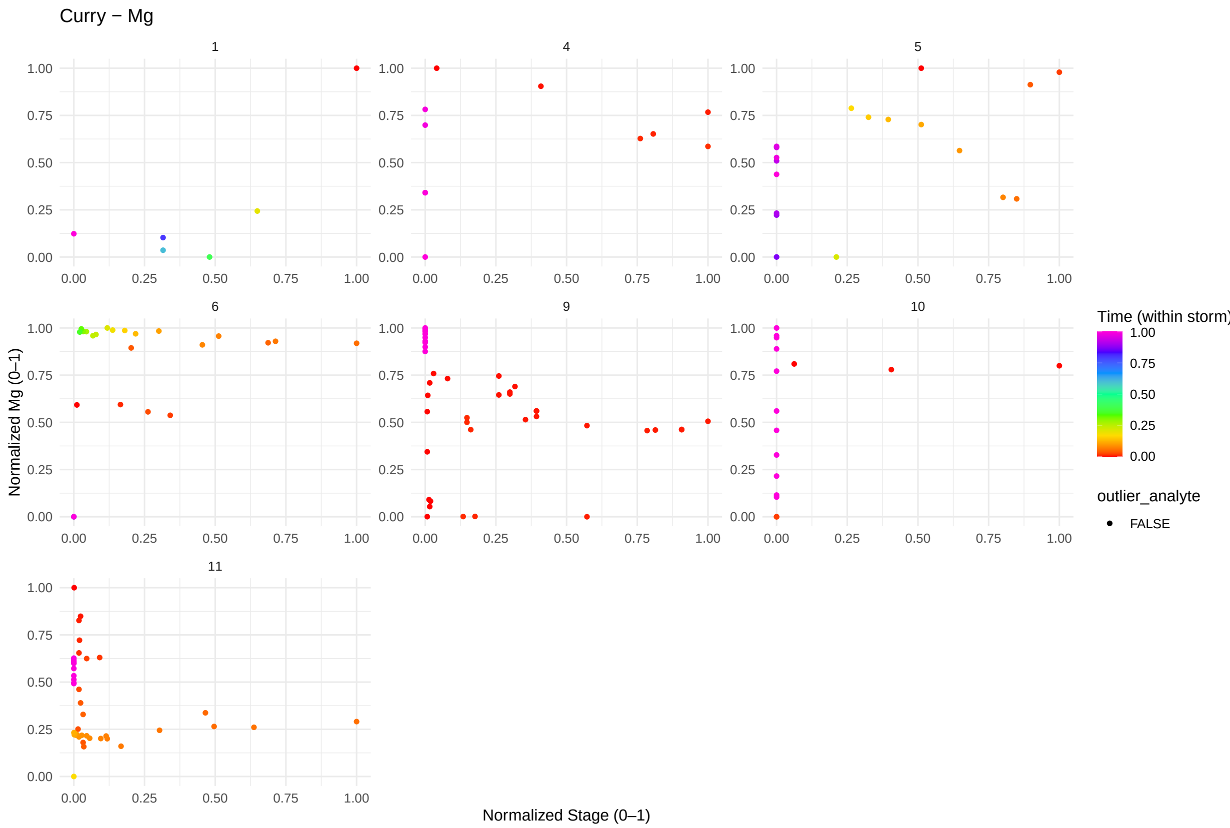
Plotting CQ

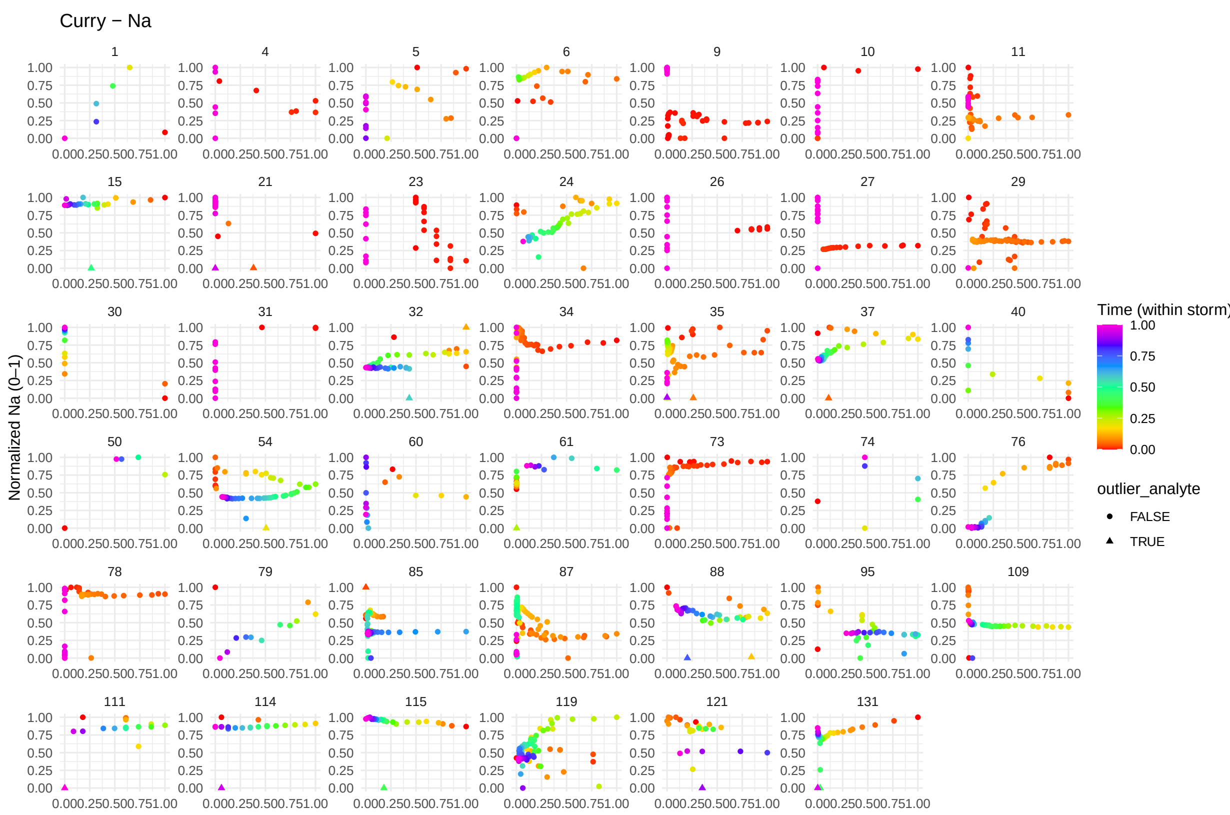

Curry − Ca
Normalized Ca (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Curry − Cl
Normalized Cl (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Curry − DOC
Normalized DOC (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Curry − K
Normalized K (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Curry − Mg
1
4
5
6
9
10
11
Normalized Mg (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE


Curry − Na
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31
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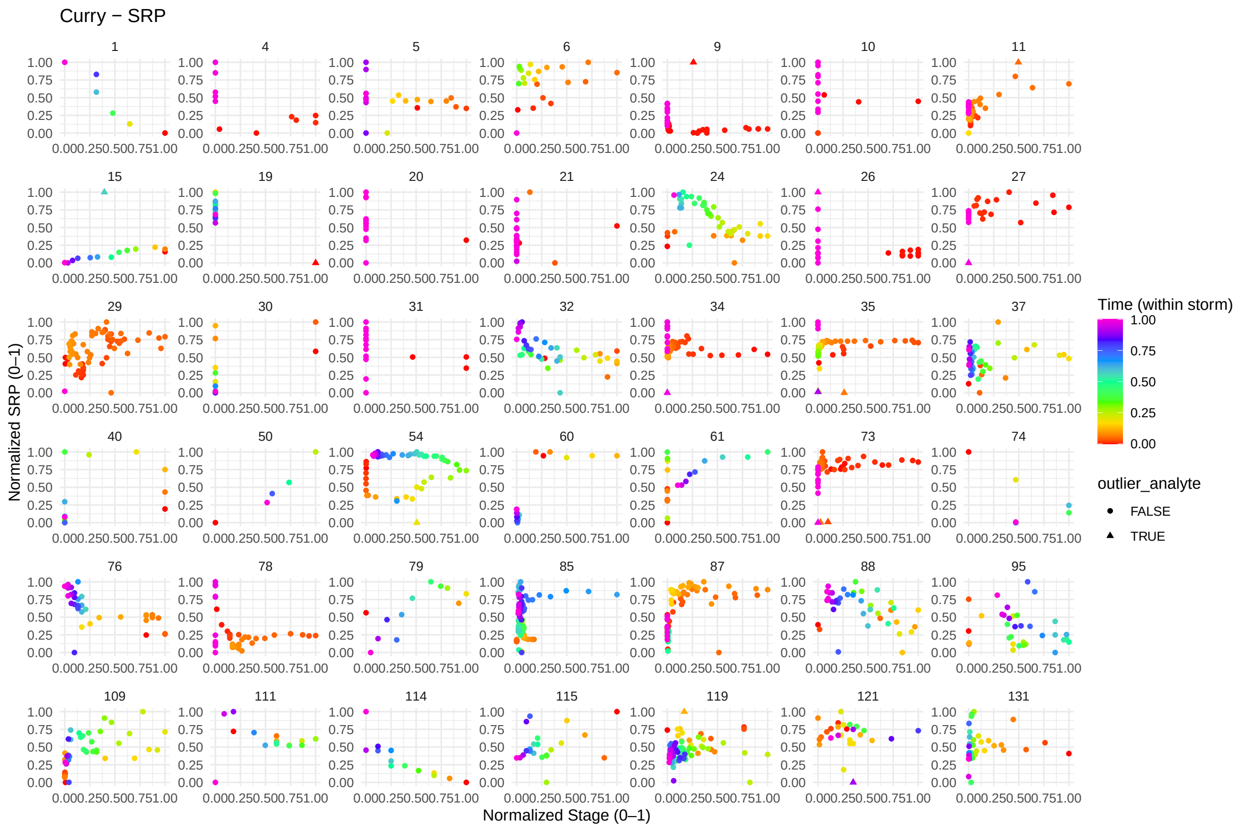
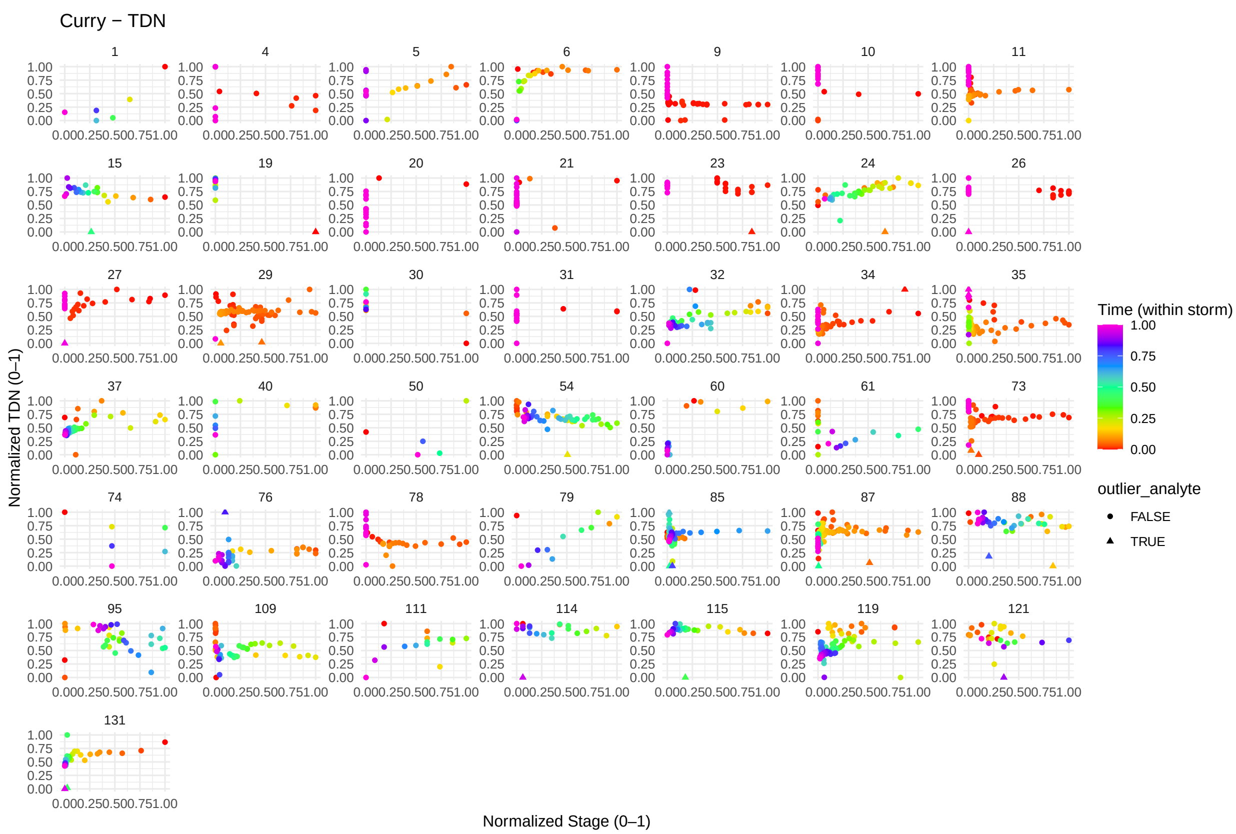
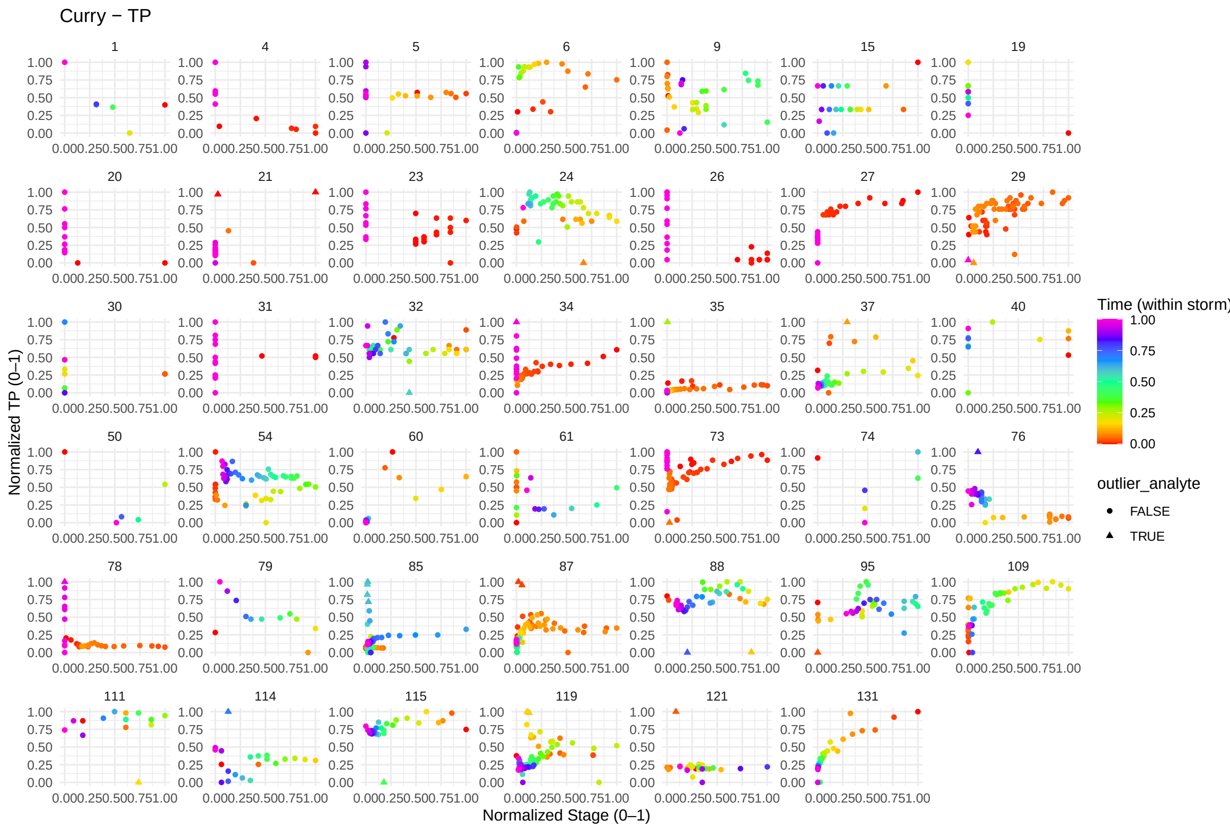
34
35
37
40
50
54
60
61
73
74
76
78
79
85
87
88
95
109
111
114
115
119
121
131
Normalized Na (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Curry − SRP
Normalized SRP (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Curry − TDN
Normalized TDN (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Curry − TP
1
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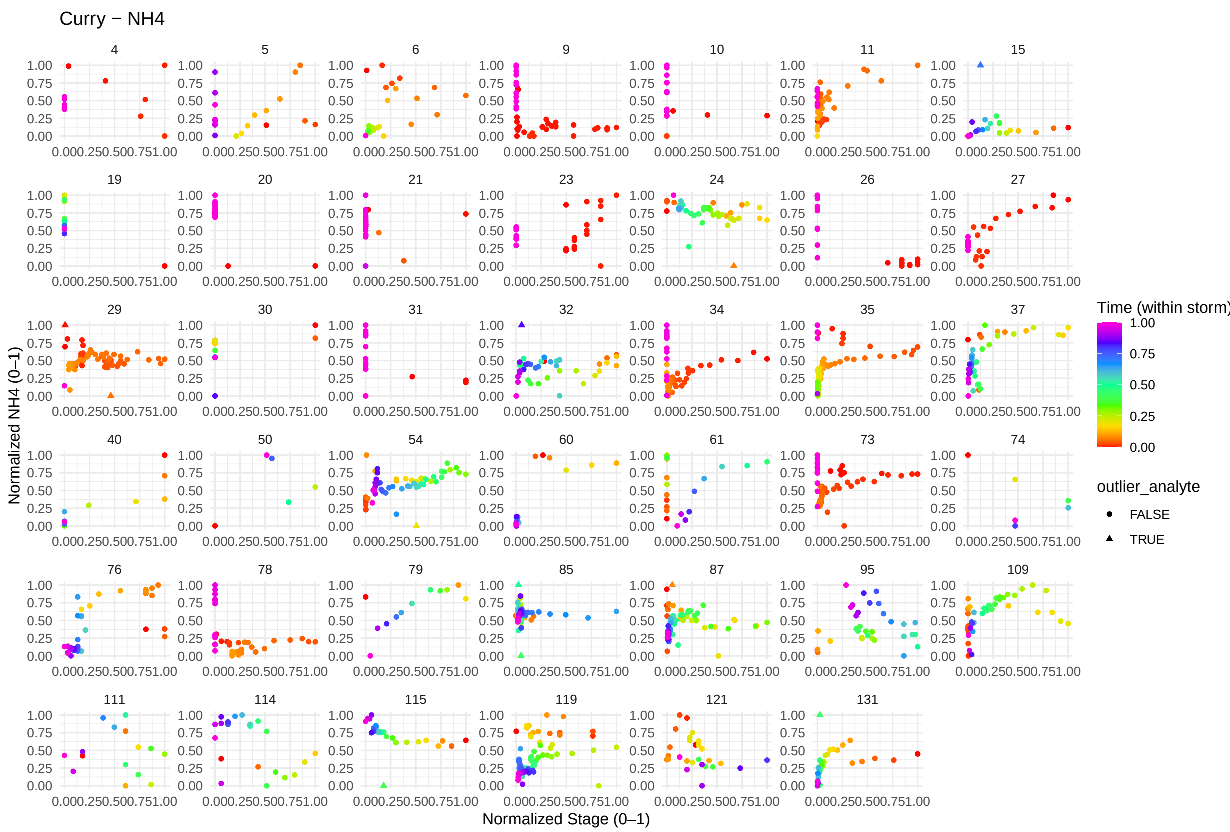
34
35
37
40
50
54
60
61
73
74
76
78
79
85
87
88
95
109
111
114
115
119
121
131
Normalized TP (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Curry − NH4
4
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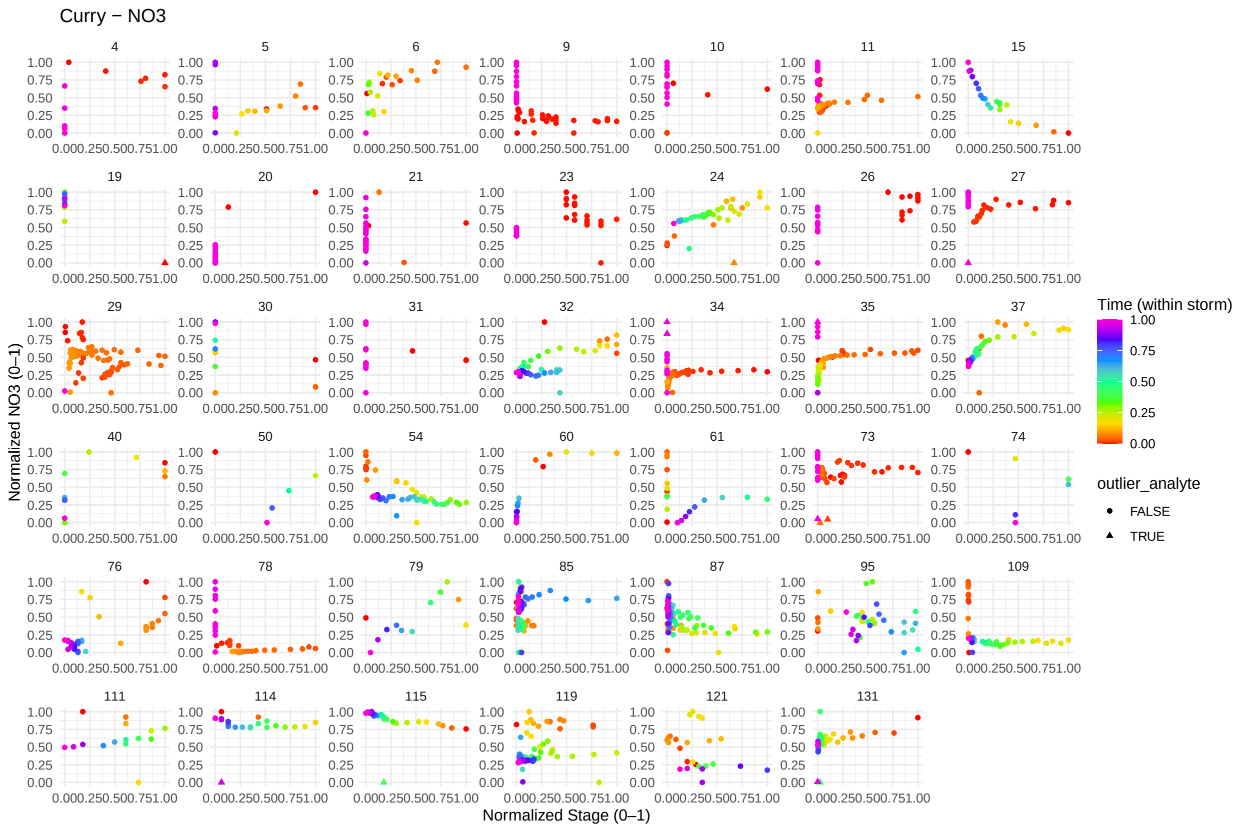
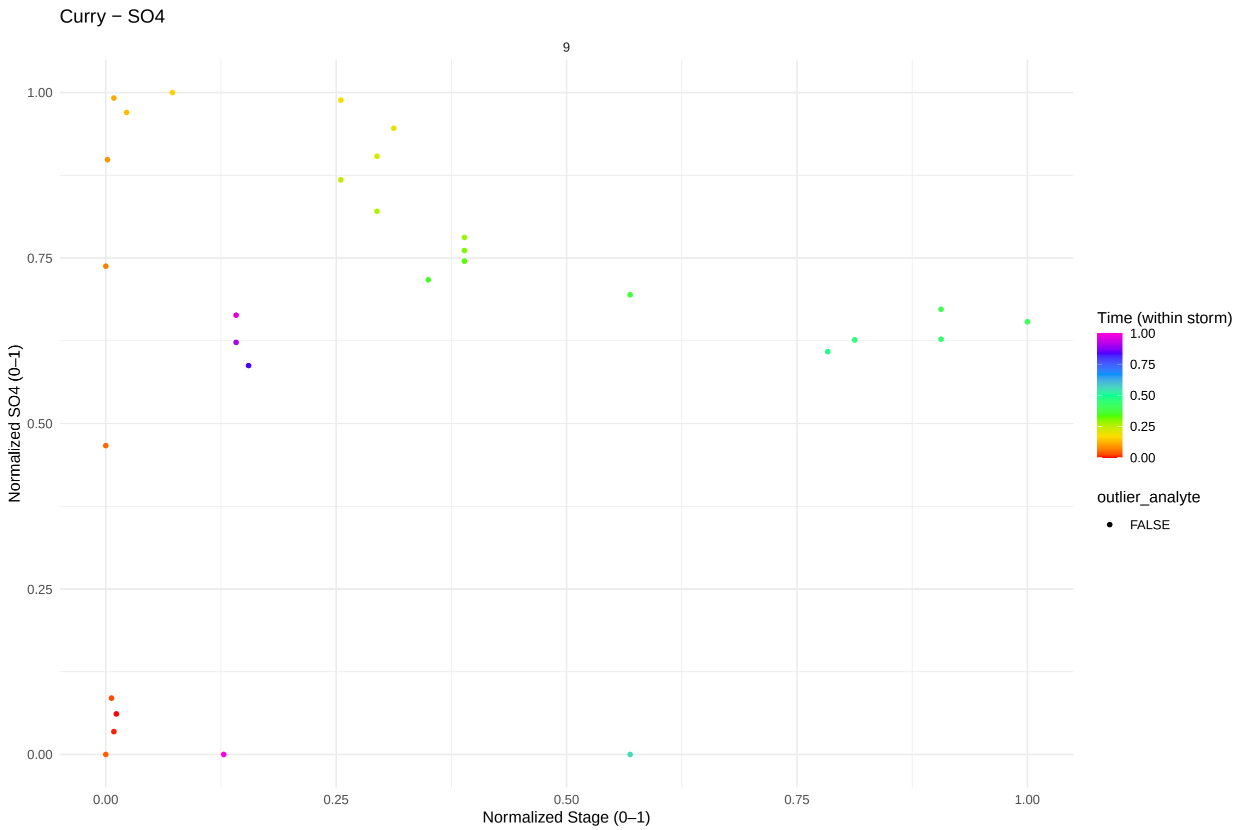
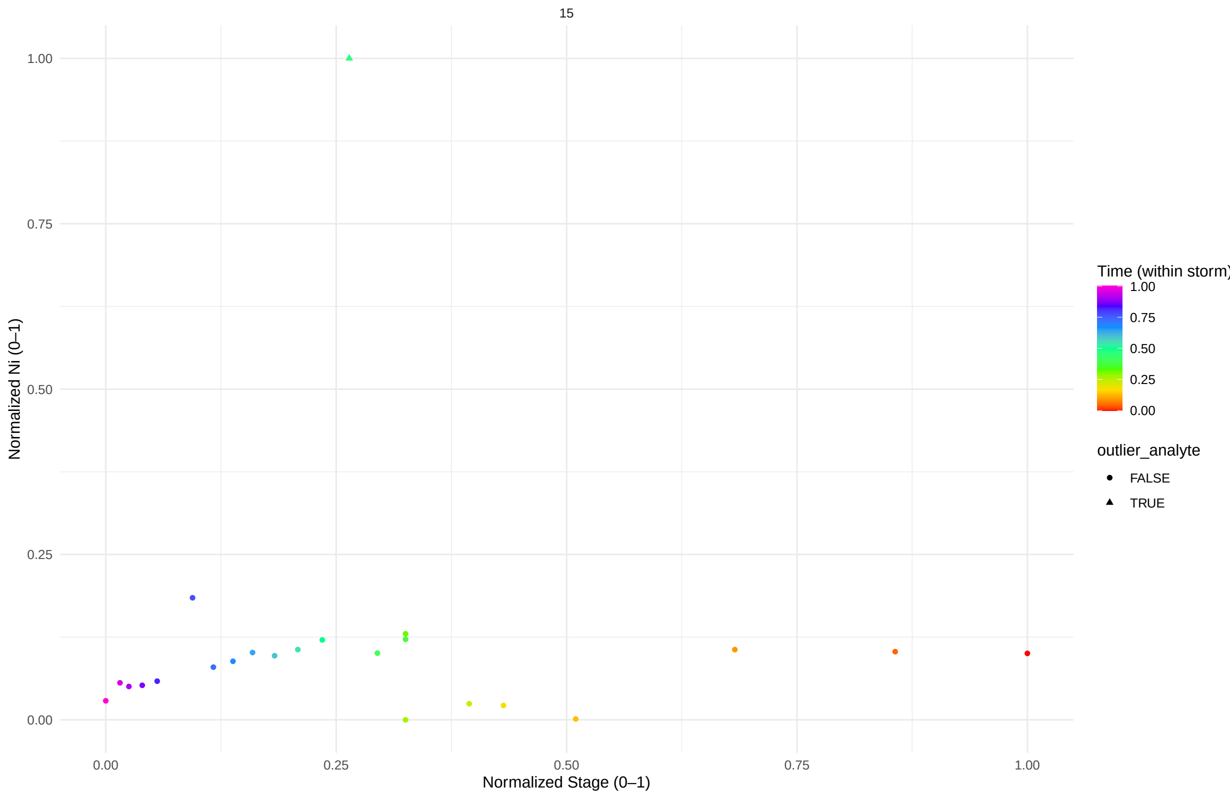
32
34
35
37
40
50
54
60
61
73
74
76
78
79
85
87
95
109
111
114
115
119
121
131
Normalized NH4 (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE

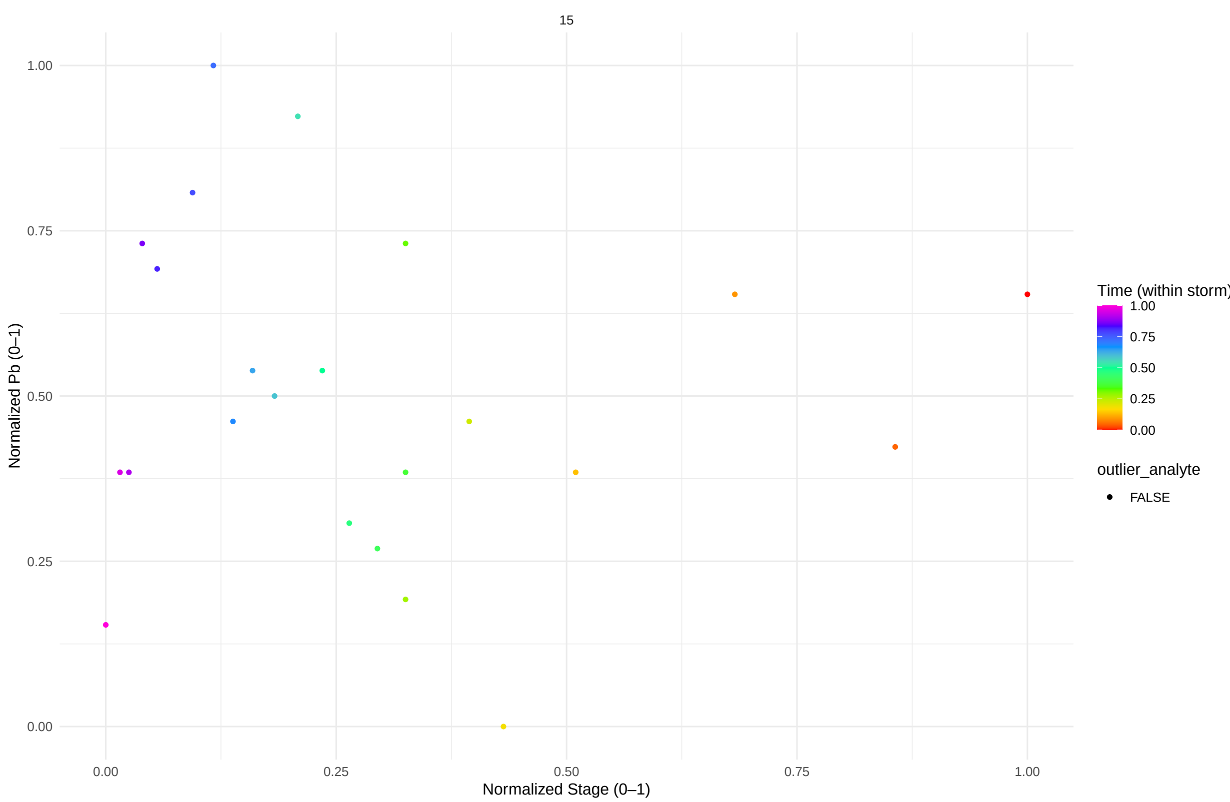

Curry − NO3
Normalized NO3 (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE

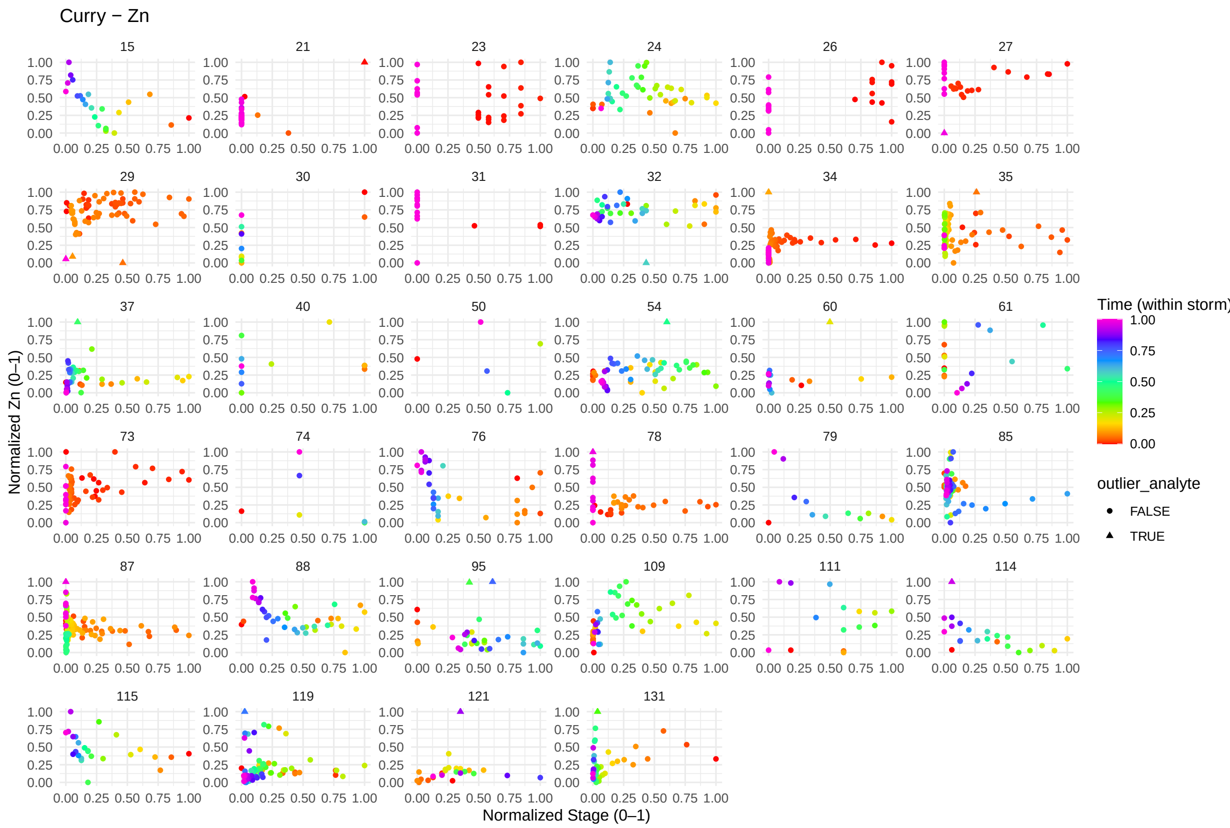

Curry − SO4
Normalized SO4 (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE

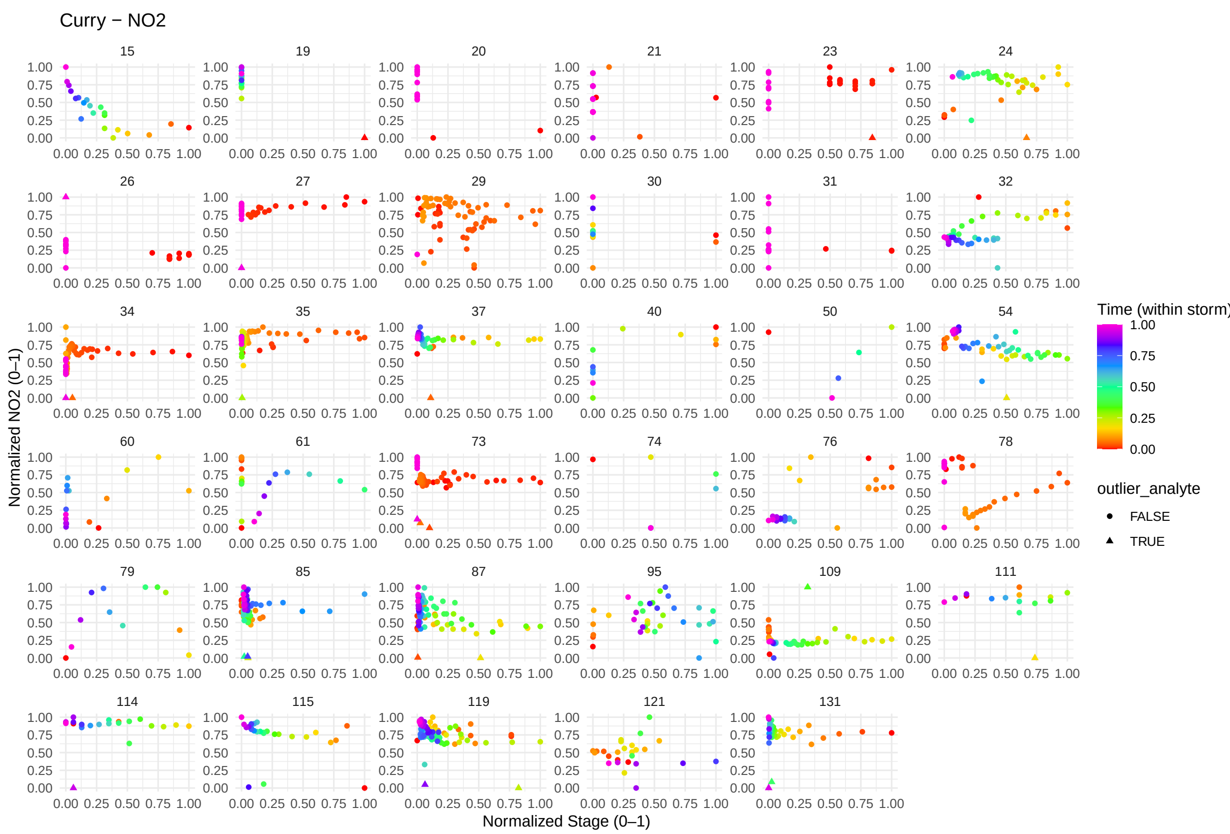
Curry − Ni

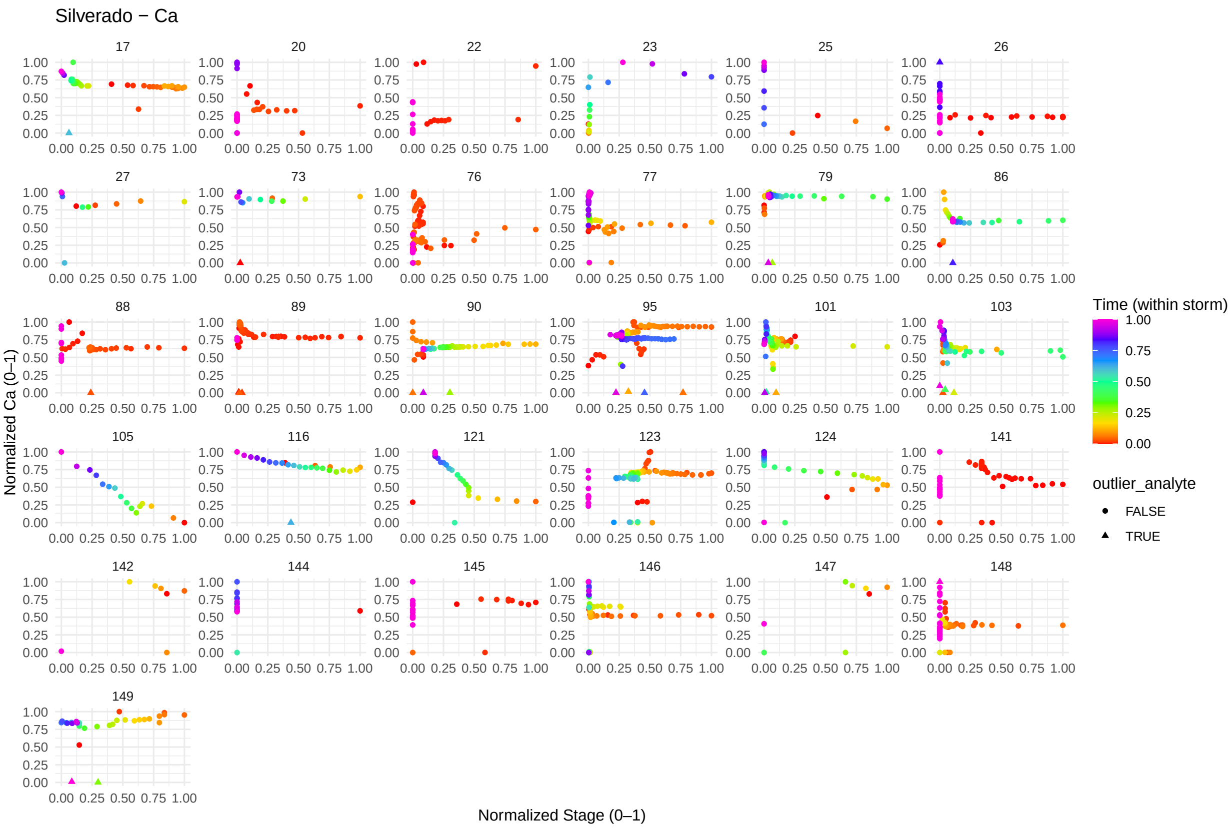


Curry − Pb




Curry − Zn
15 21 23 24 26 27 29 30 31 32 34 35 37 40 50 54 60 61 73 74 76 78 79 85 87 88 95 109 111 114 115 119 121 131
Normalized Zn (0–1)
Normalized Stage (0–1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00
outlier_analyte
FALSE
TRUE


Curry − NO2
15 19 20 21 23 24 26 27 29 30 31 32 34 35 37 40 50 54 60 61 73 74 76 78 79 85 87 95 109 111 114 115 119 121 131
Normalized NO2 (0–1)
Normalized Stage (0–1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00
outlier_analyte
FALSE
TRUE


Silverado − Ca
17
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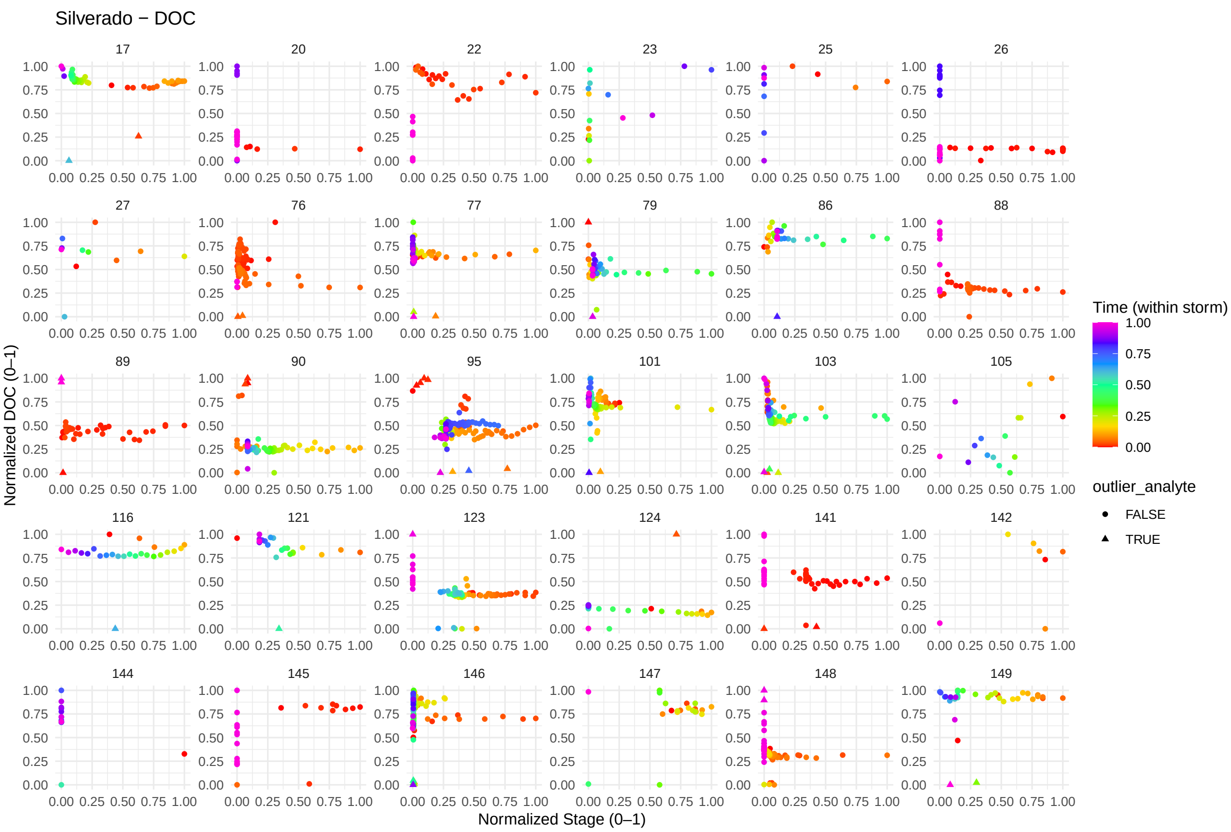
101
103
105
116
121
123
124
141
142
144
145
146
147
148
149
Normalized Ca (0–1)
Normalized Stage (0–1)
0.00 0.25 0.50 0.75 1.00
Time (within storm)
1.00
0.75
0.50
0.25
0.00
outlier_analyte
FALSE
TRUE


Silverado − DOC
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86
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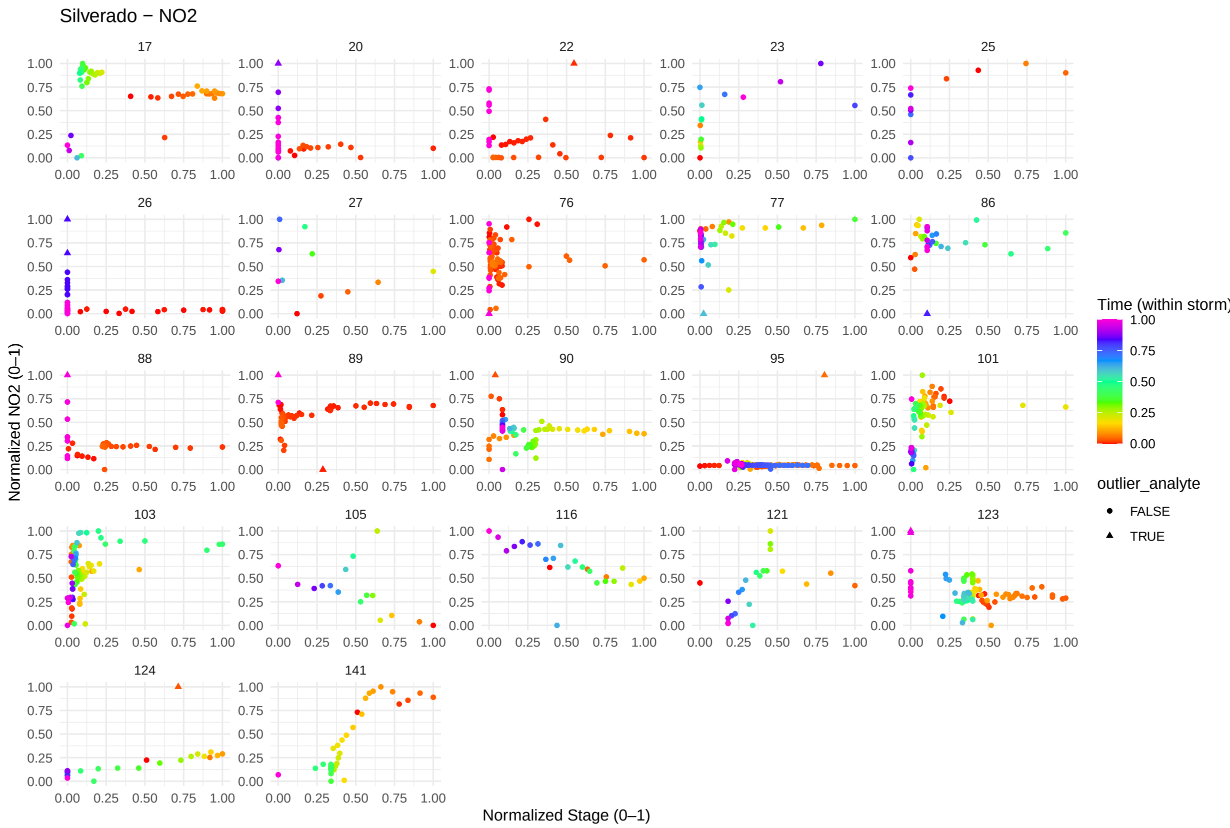
89
90
95
101
103
105
116
121
123
124
141
142
144
145
146
147
148
149
Normalized DOC (0–1)
Normalized Stage (0–1)
0.00 0.25 0.50 0.75 1.00
Time (within storm)
1.00
0.75
0.50
0.25
0.00
outlier_analyte
FALSE
TRUE


Silverado − NO2
17
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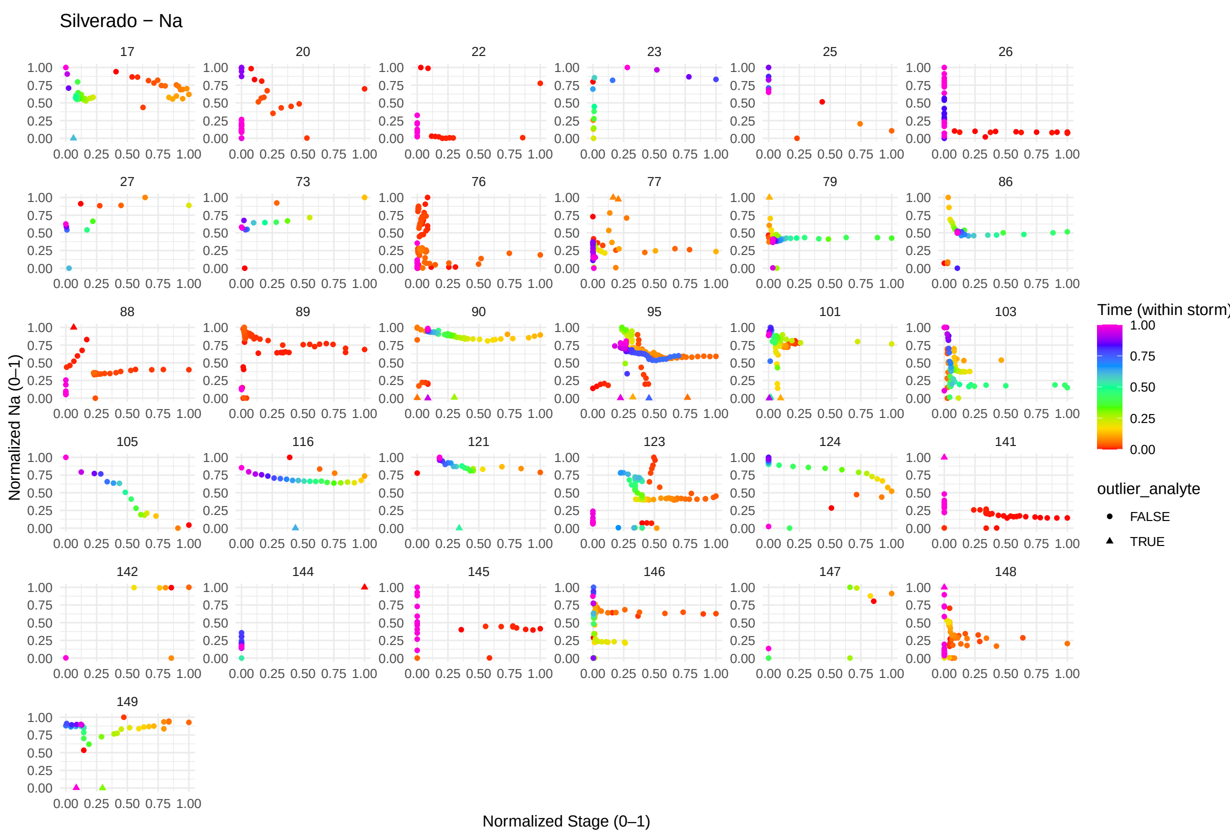
22
23
25
26
27
76
77
86
88
89
90
95
101
103
105
116
121
123
124
141
Normalized NO2 (0–1)
Normalized Stage (0–1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00
outlier_analyte
FALSE
TRUE


Silverado − Na
17
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27
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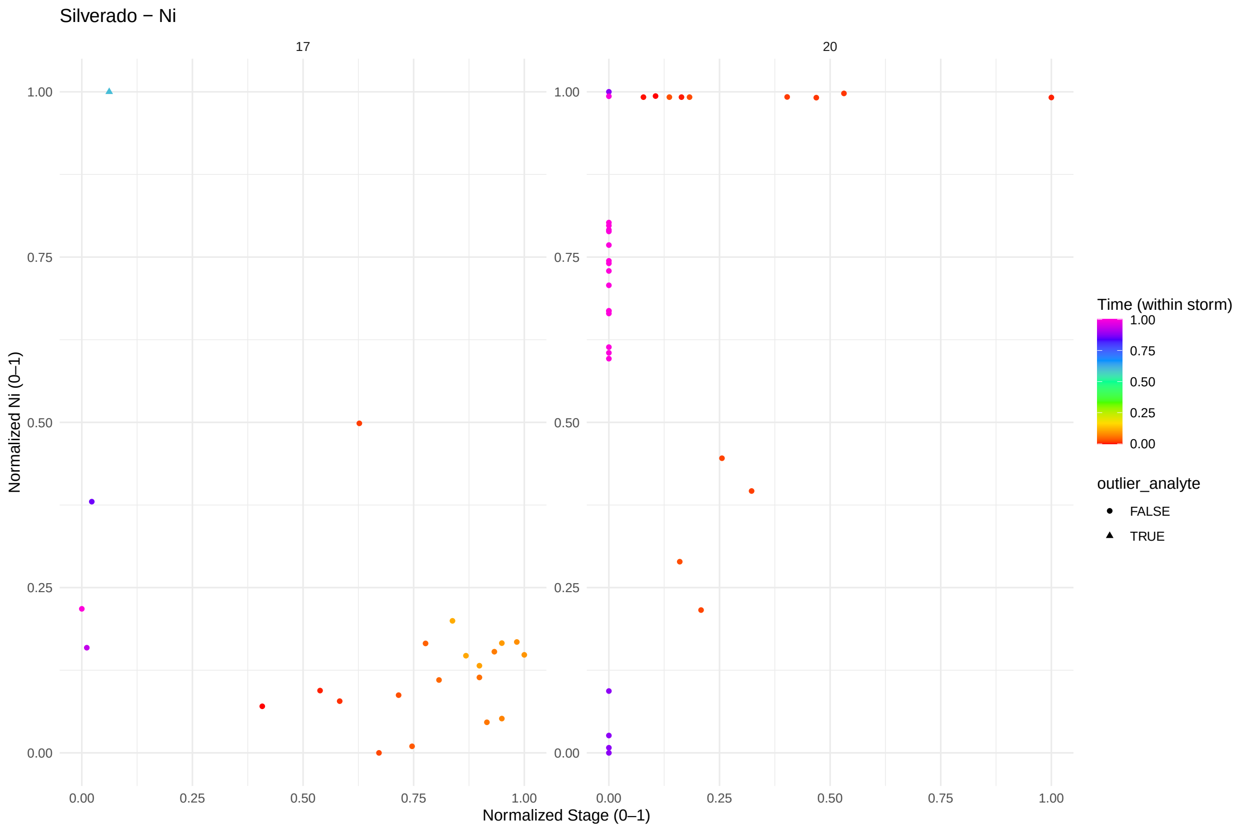
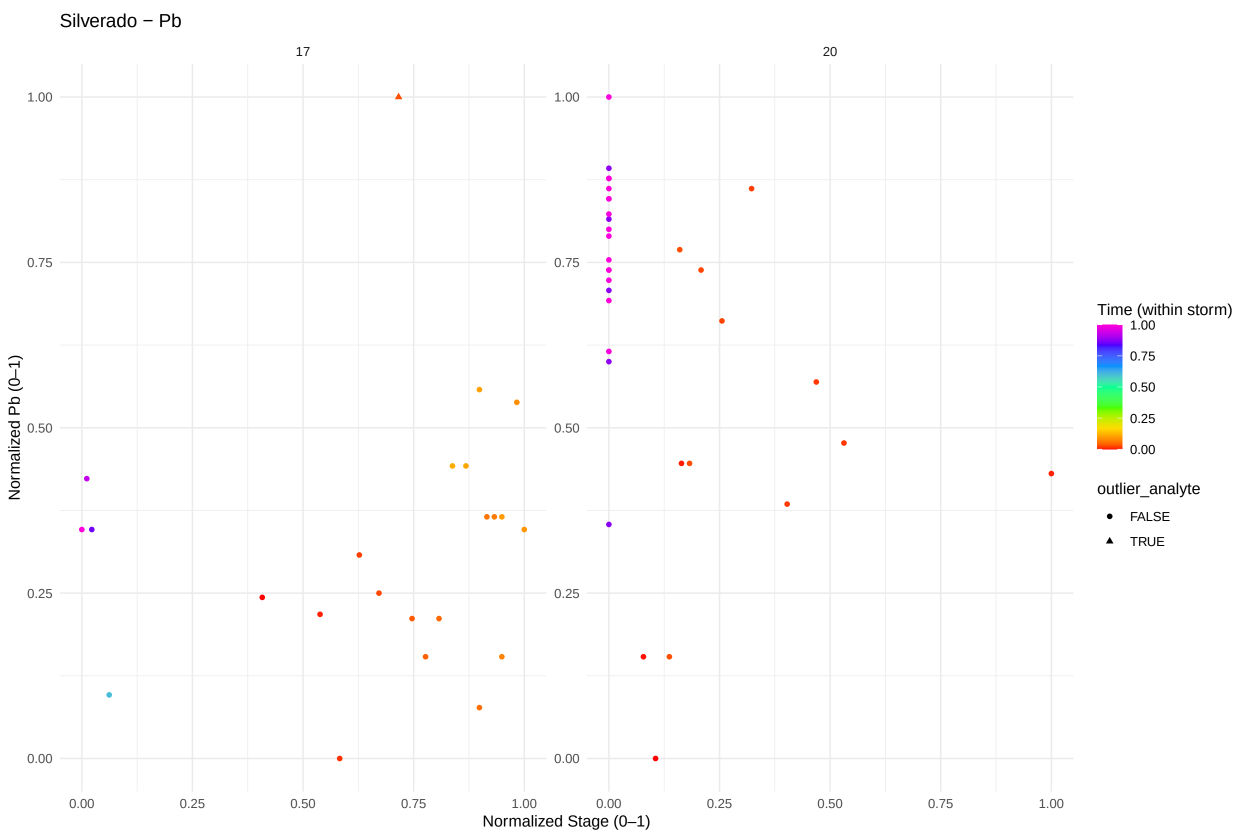
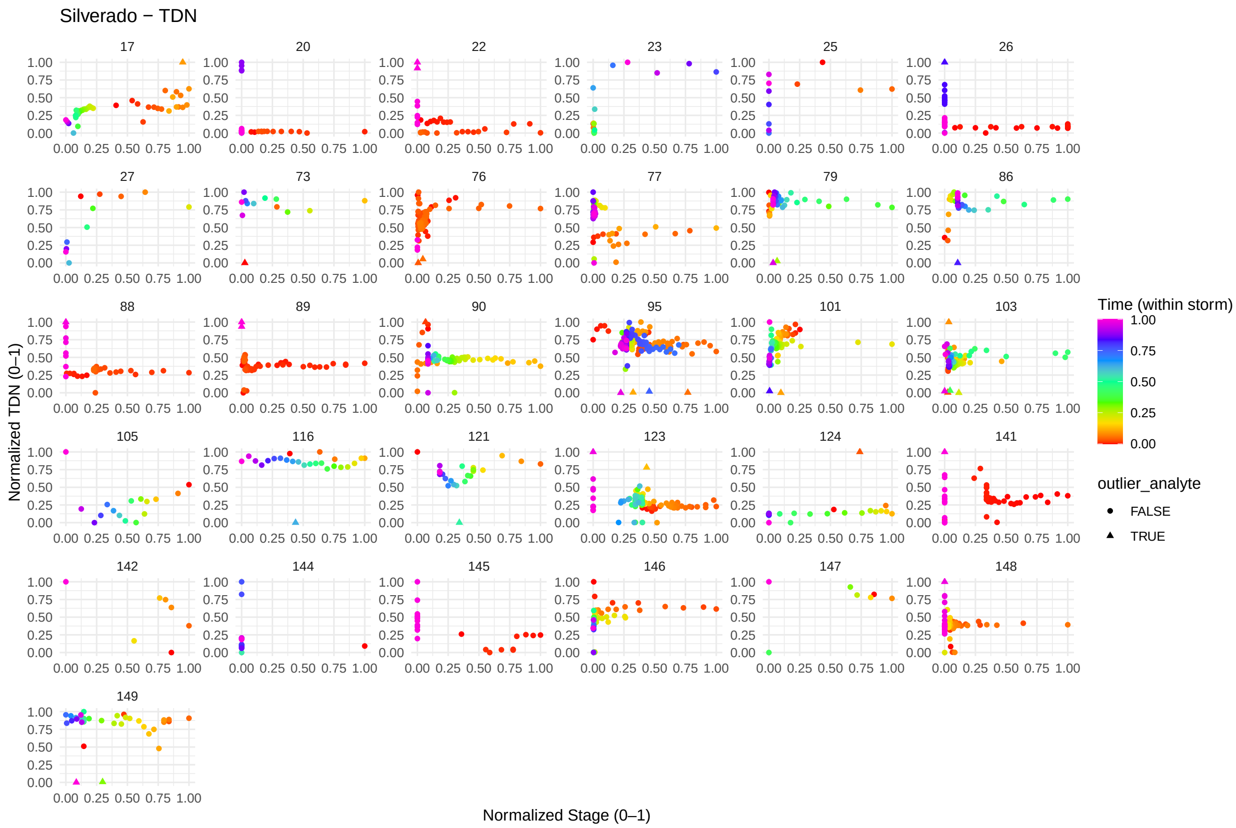
77
79
86
88
89
90
95
101
103
105
116
121
123
124
141
142
144
145
146
147
148
149
Normalized Na (0–1)
Normalized Stage (0–1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00
outlier_analyte
FALSE
TRUE



Silverado − Ni
17
20
Normalized Ni (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE



Silverado − Pb
17
20
Normalized Pb (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Silverado − TDN
17
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22
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27
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79
86
88
89
90
95
101
103
105
116
121
123
124
141
142
144
145
146
147
148
149
Normalized TDN (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Silverado − TP
17
20
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23
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26
27
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79
86
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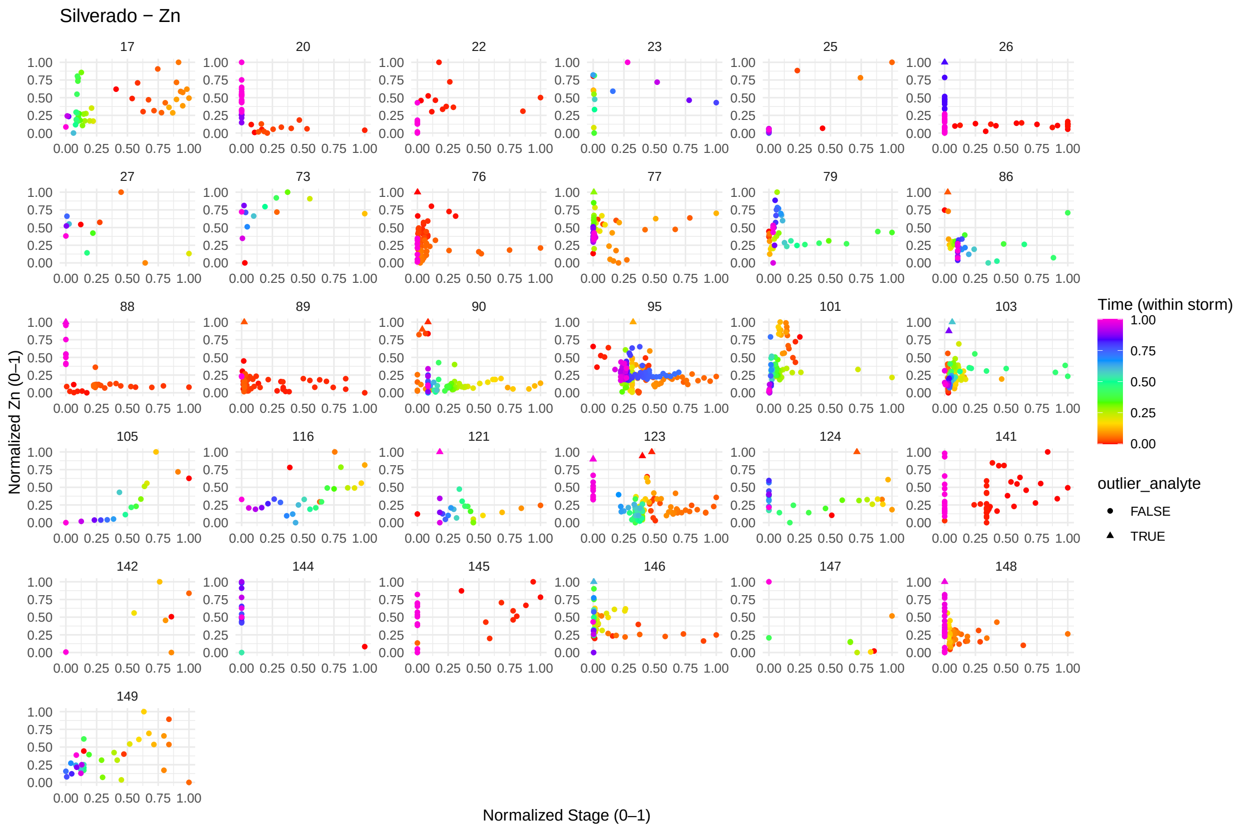
89
90
95
101
103
116
121
123
124
141
142
144
145
146
147
148
149
Normalized TP (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Silverado − Zn
17
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86
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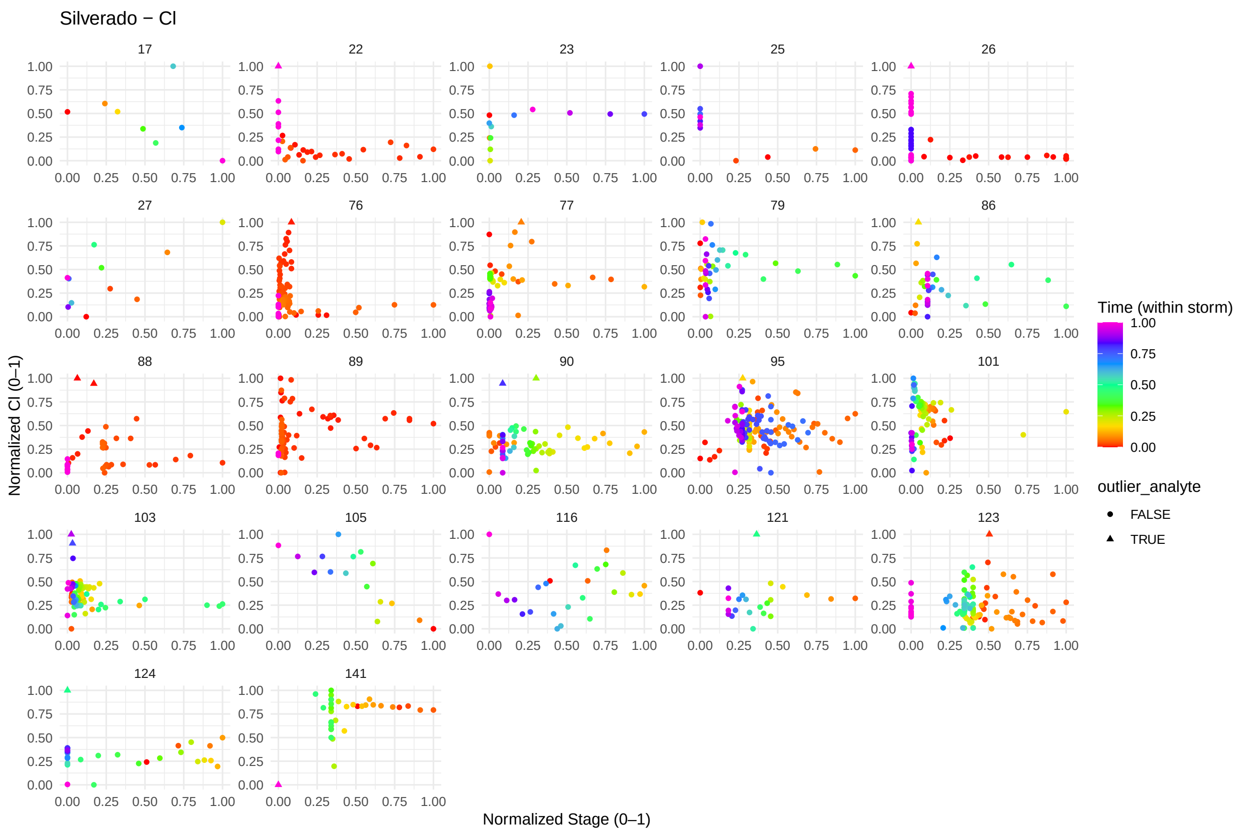
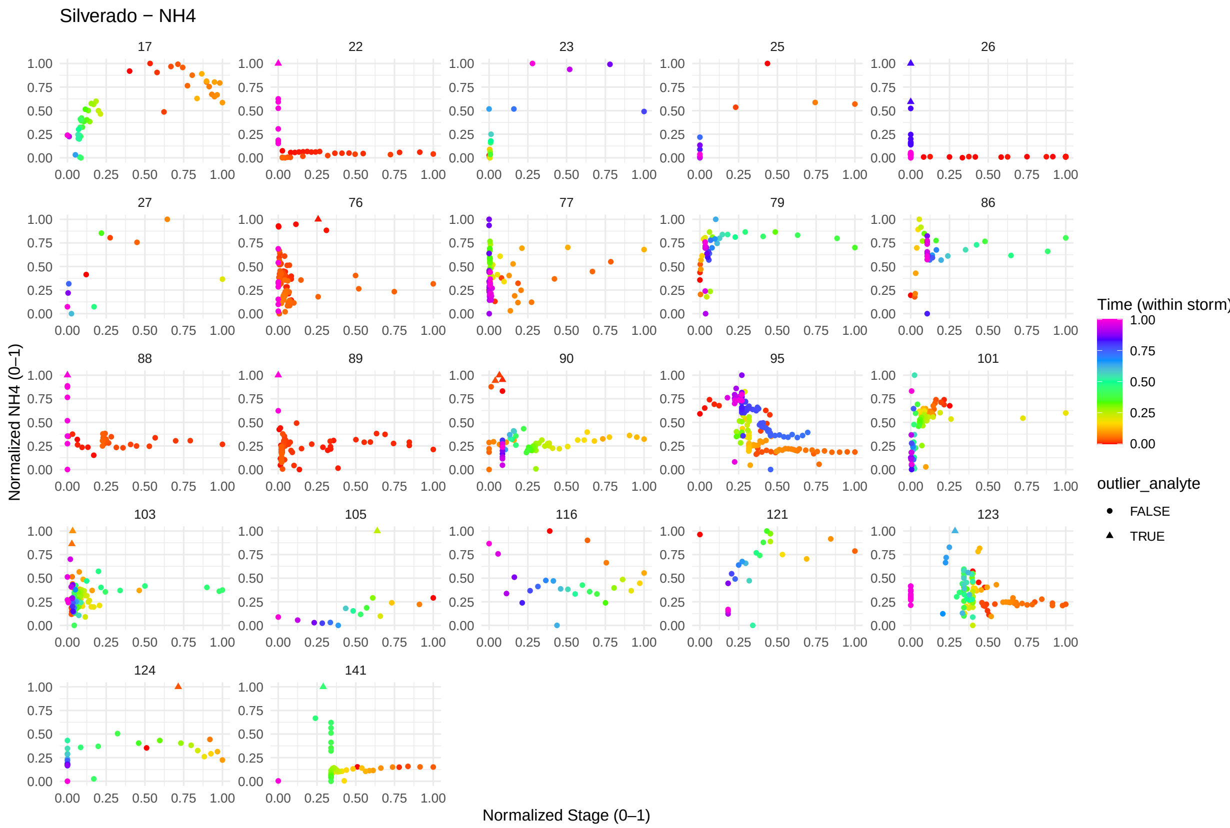
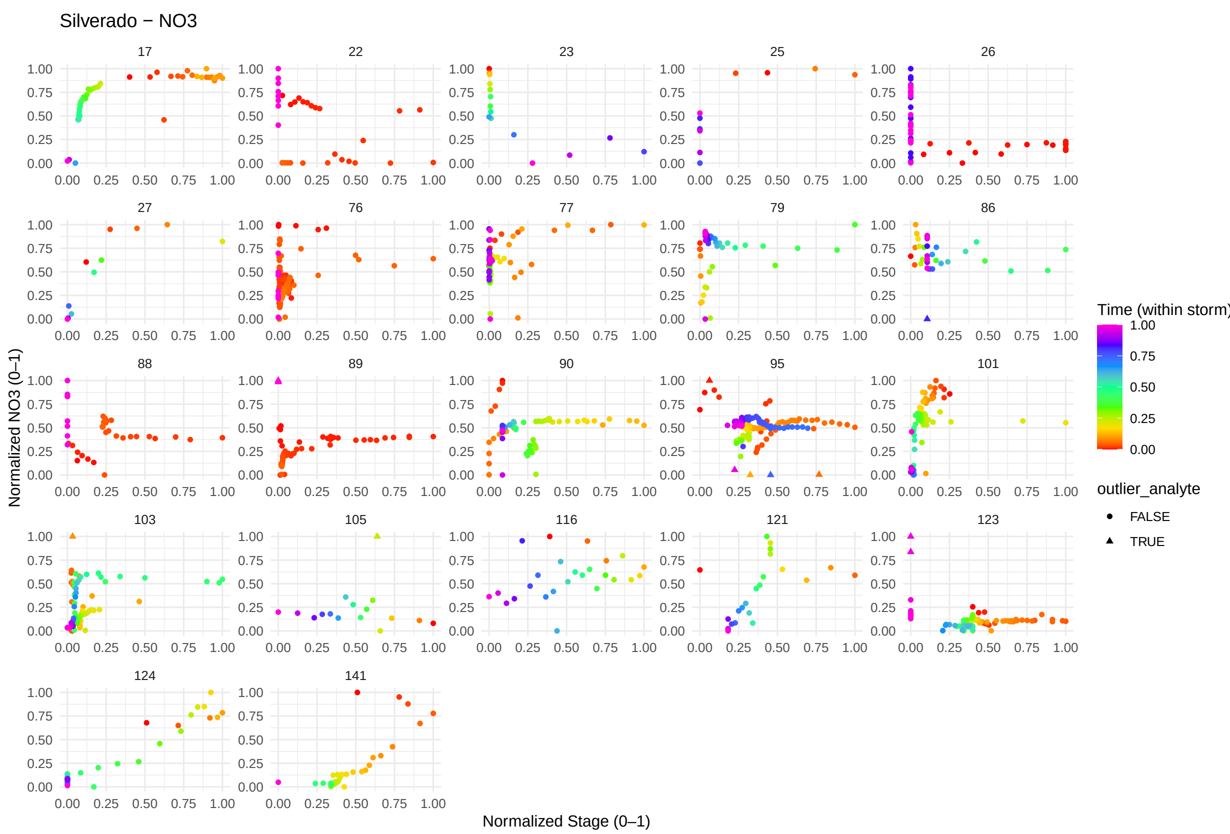
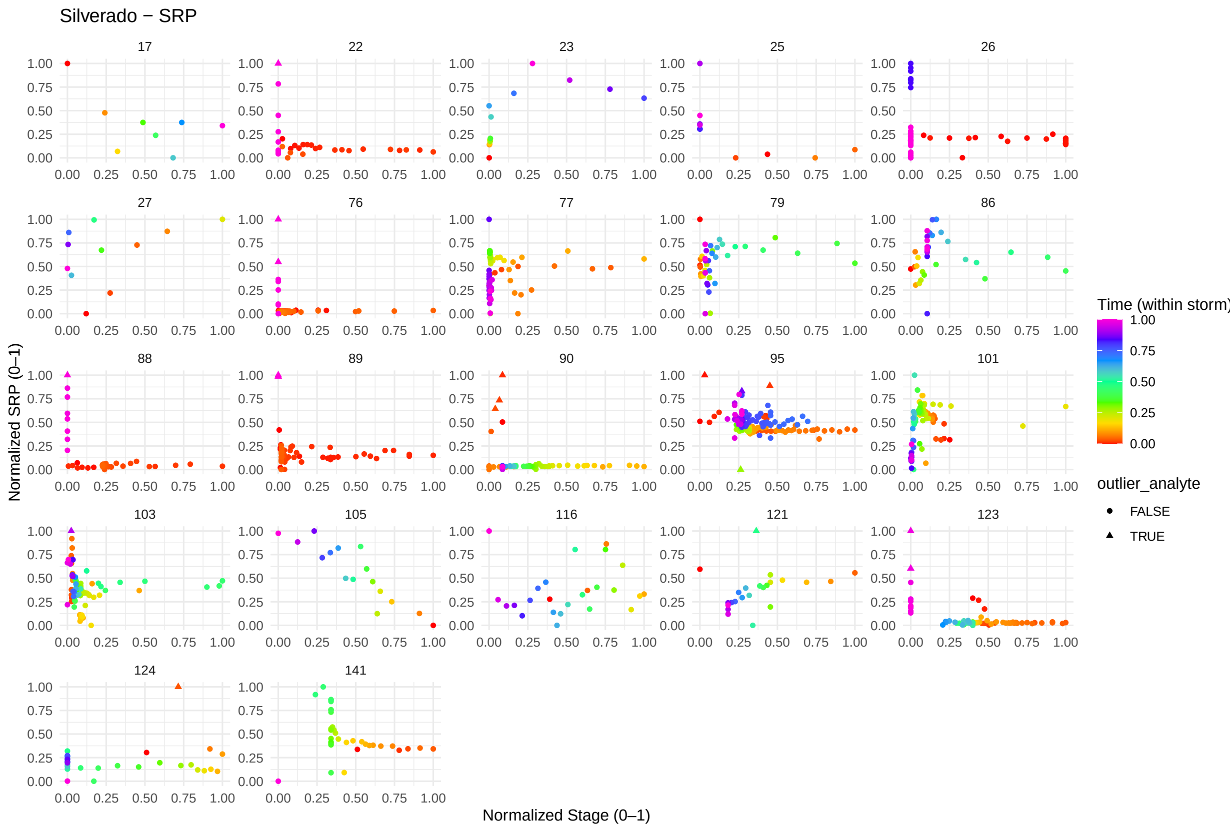
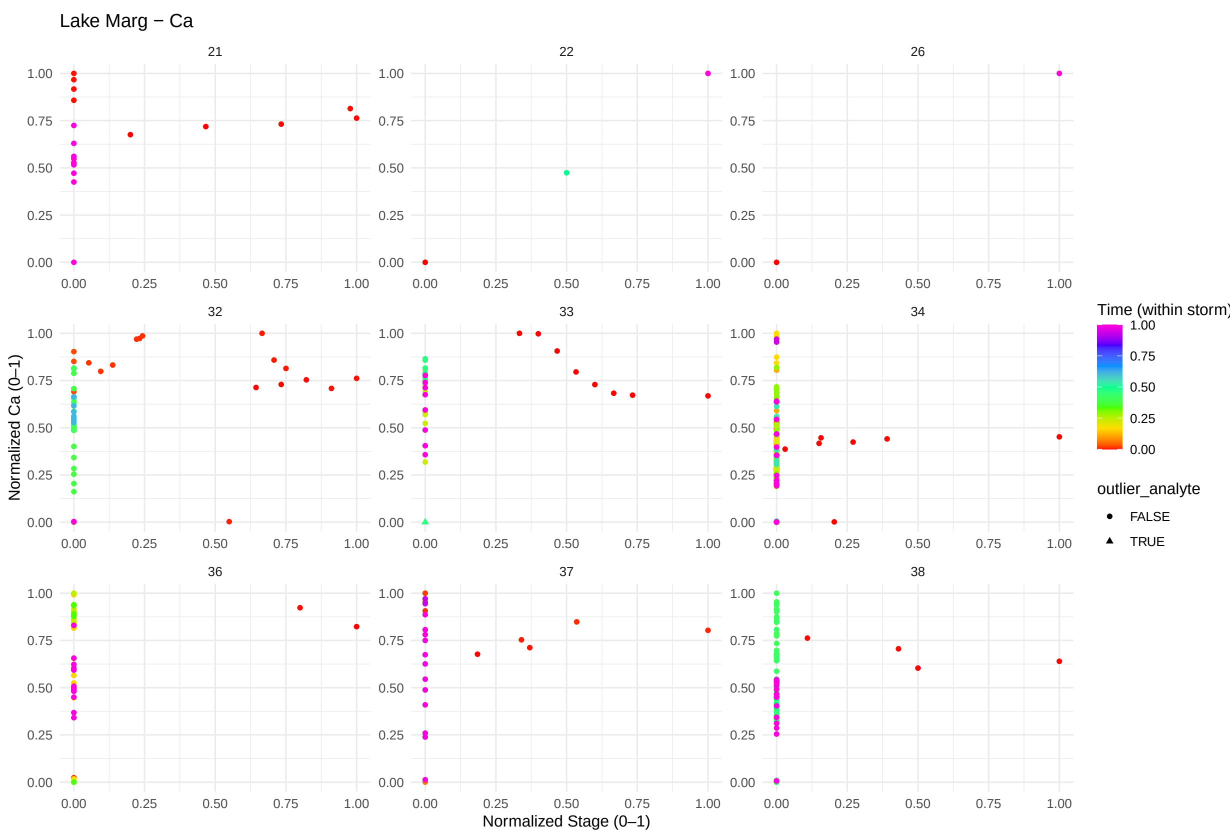
89
90
95
101
103
105
116
121
123
124
141
142
144
145
146
147
148
149
Normalized Zn (0–1)
Normalized Stage (0–1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00
outlier_analyte
FALSE
TRUE

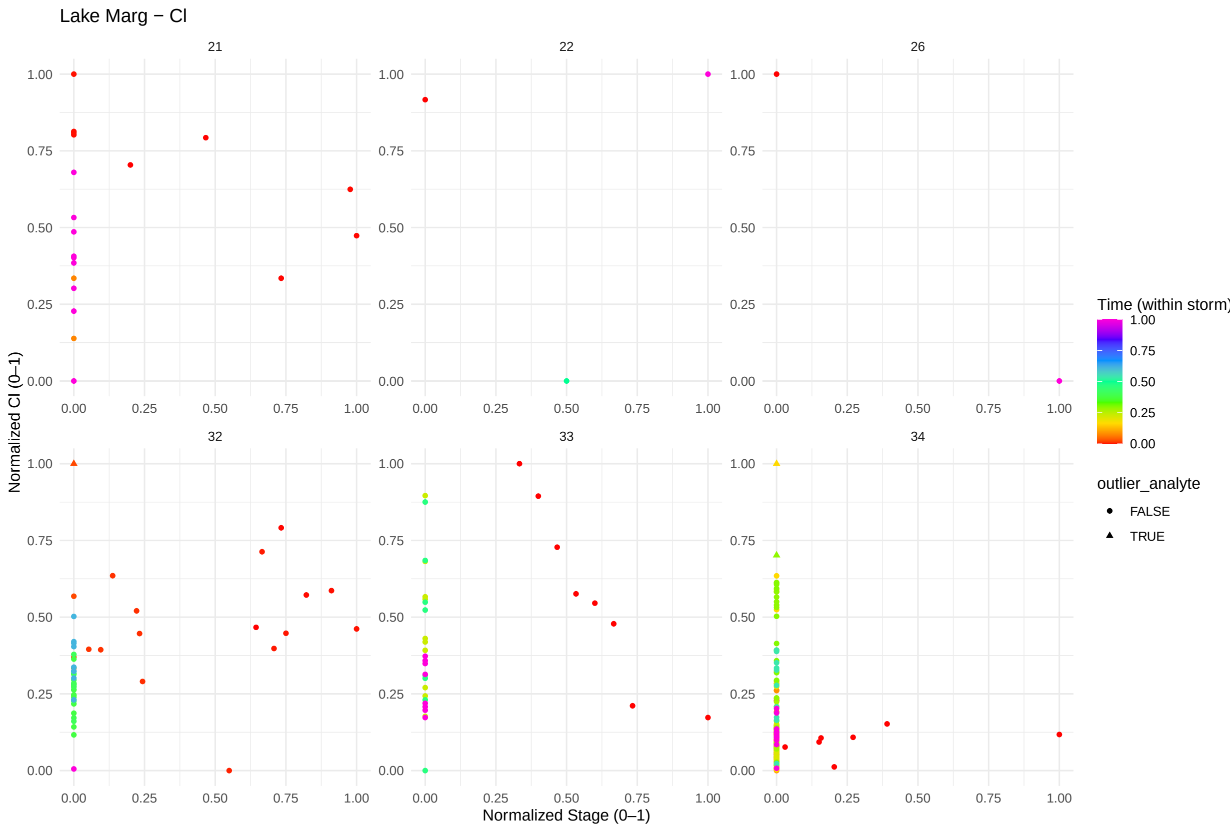

Silverado − Cl
17
22
23
25
26
27
76
77
79
86
88
89
90
95
101
103
105
116
121
123
124
141
Normalized Cl (0–1)
Normalized Stage (0–1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00
outlier_analyte
FALSE
TRUE

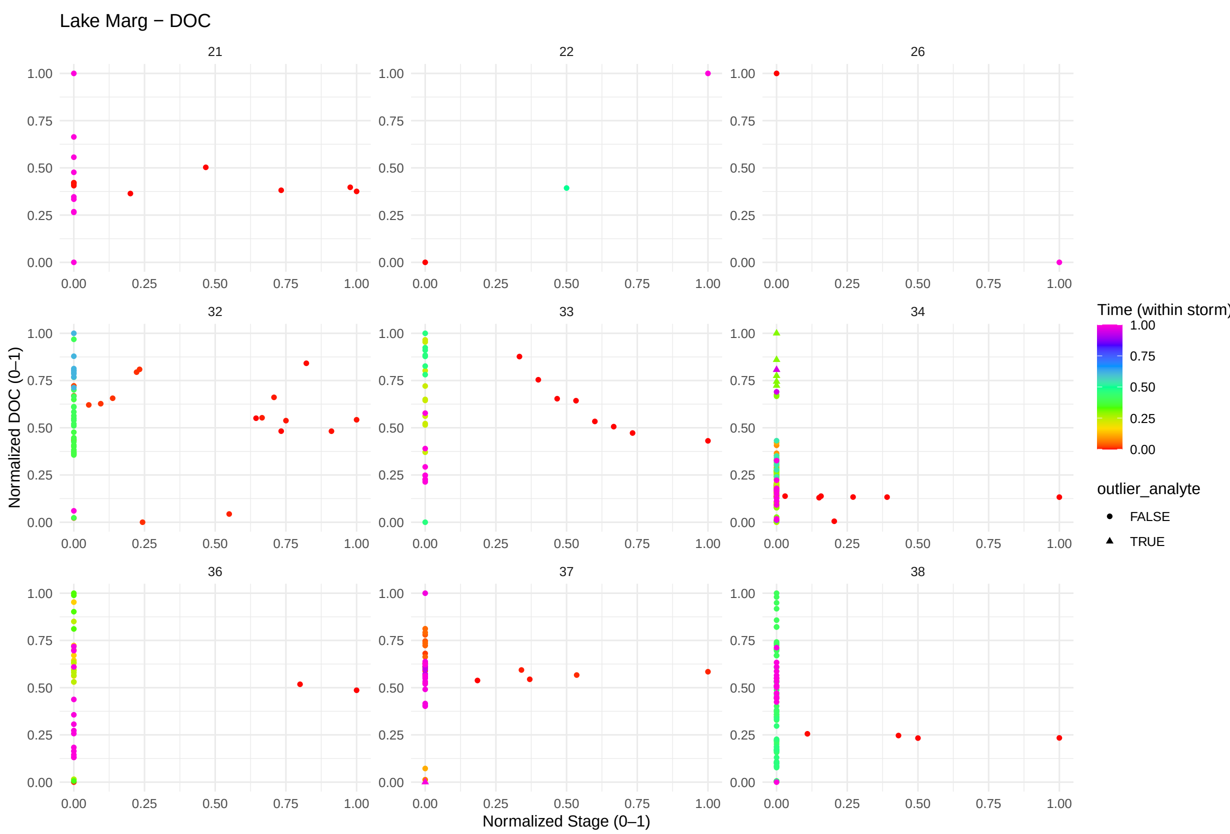

Silverado − NH4
17
22
23
25
26
27
76
77
79
86
88
89
90
95
101
103
105
116
121
123
124
141
0.00
0.25
0.50
0.75
1.00
Normalized NH4 (0–1)
Normalized Stage (0–1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00
outlier_analyte
FALSE
TRUE

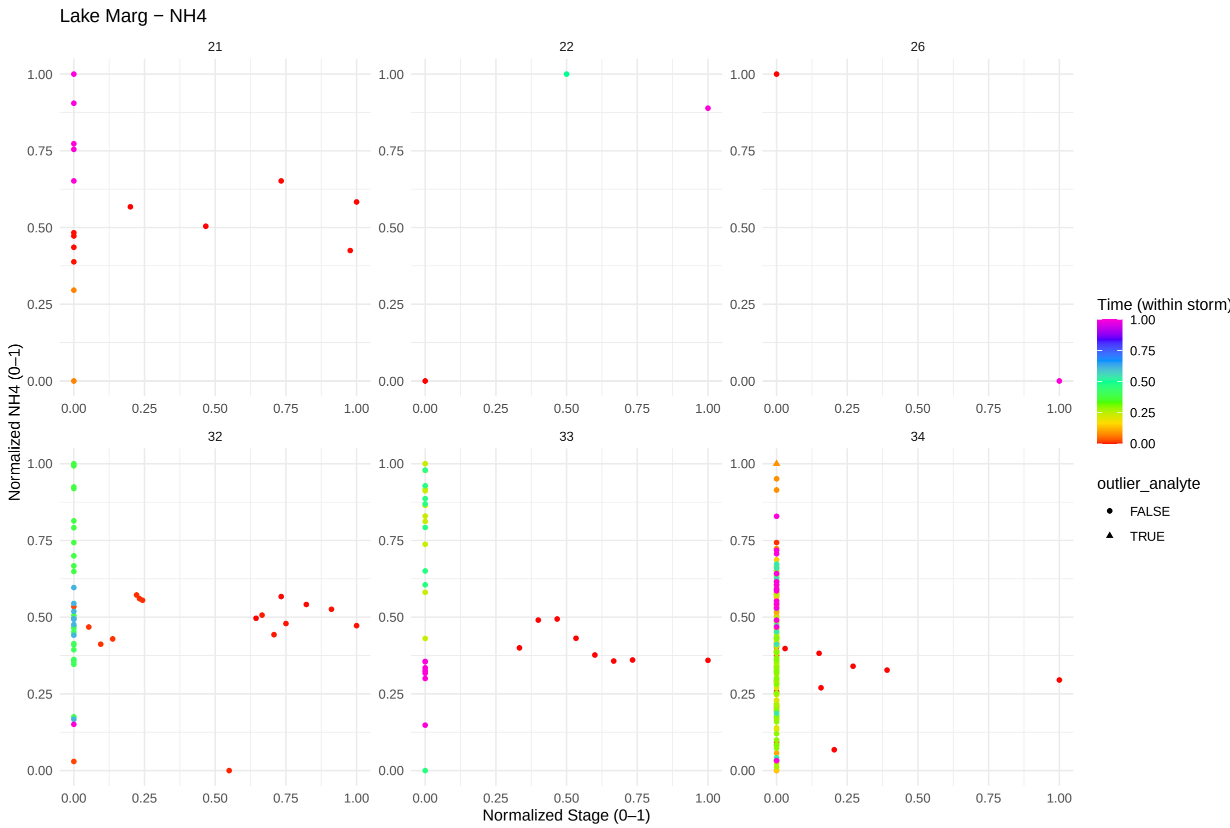

Silverado − NO3
17
22
23
25
26
27
76
77
79
86
88
89
90
95
101
103
105
116
121
123
124
141
0.00
0.25
0.50
0.75
1.00
Normalized NO3 (0–1)
Normalized Stage (0–1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00
outlier_analyte
FALSE
TRUE

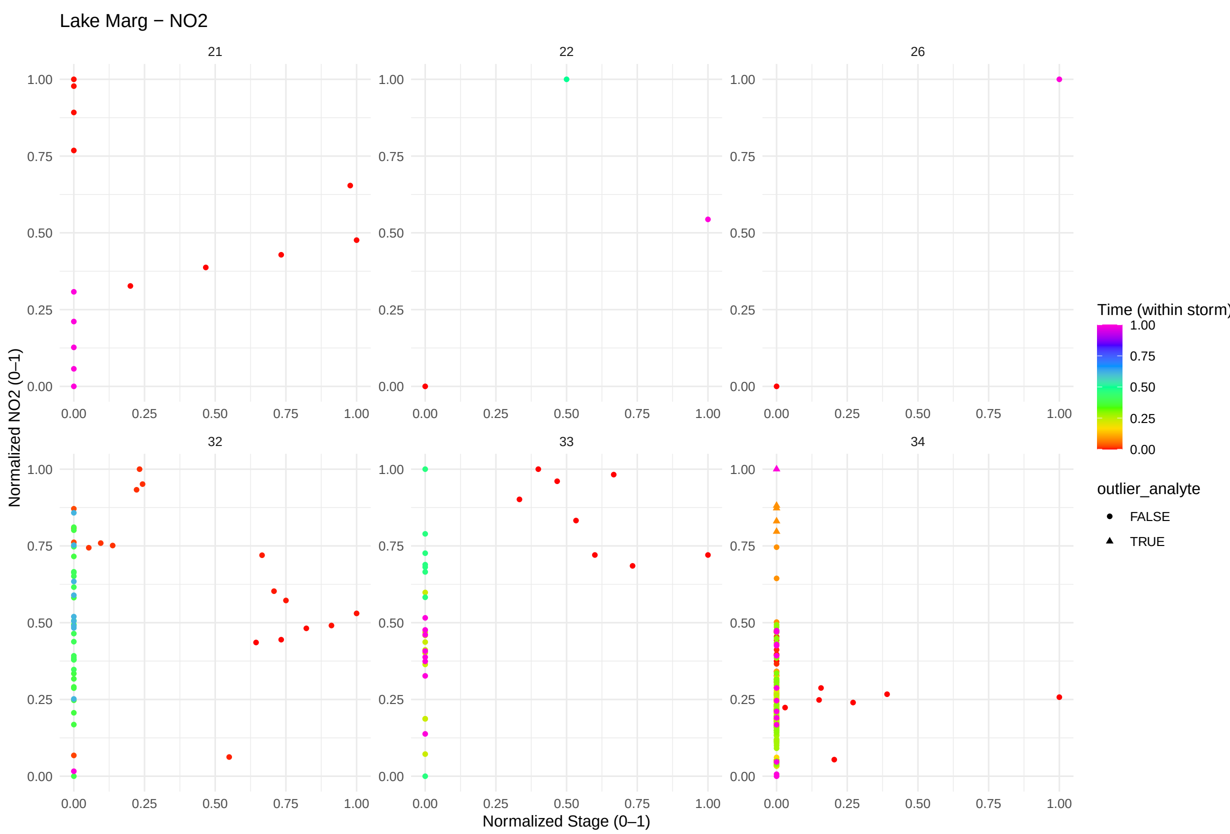

Silverado − SRP
Normalized SRP (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE

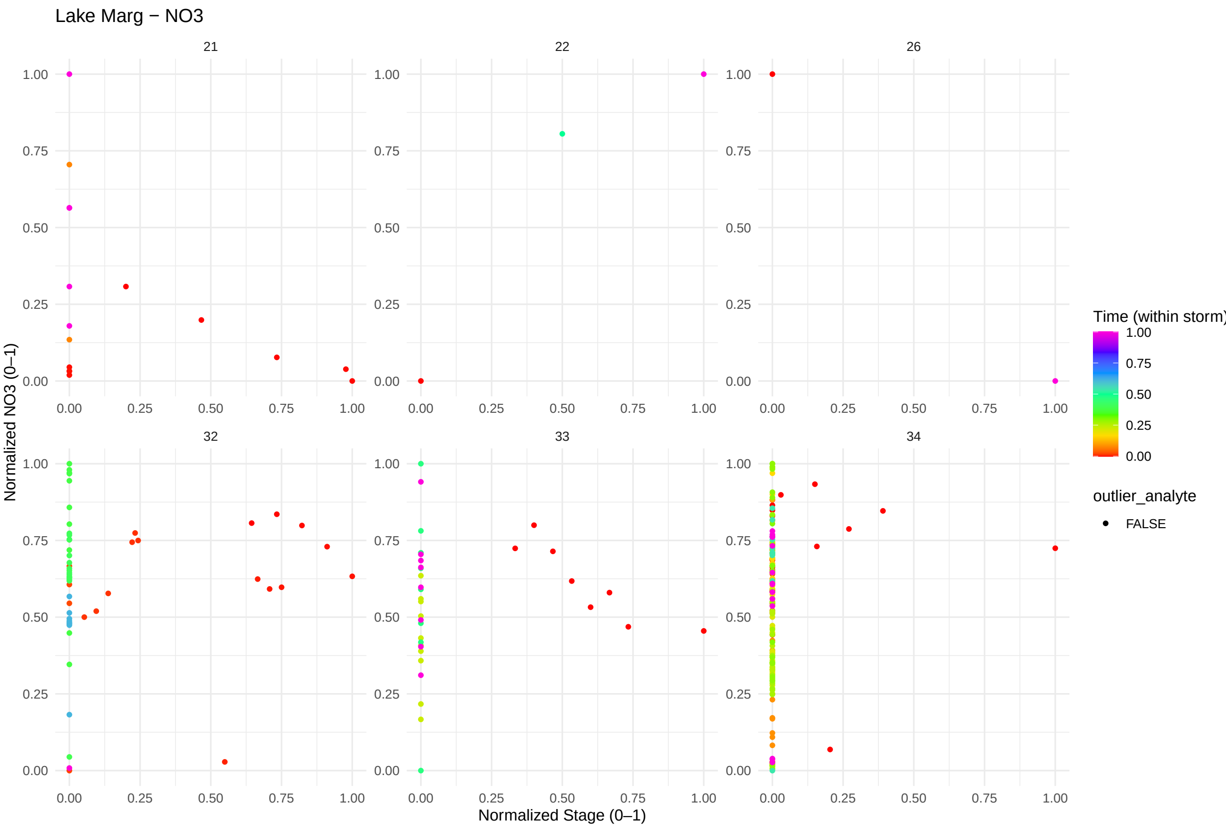

Lake Marg − Ca
Normalized Ca (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Lake Marg − Cl
21
22
26
32
33
34
Normalized Cl (0–1)
Normalized Stage (0–1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00
outlier_analyte
FALSE
TRUE

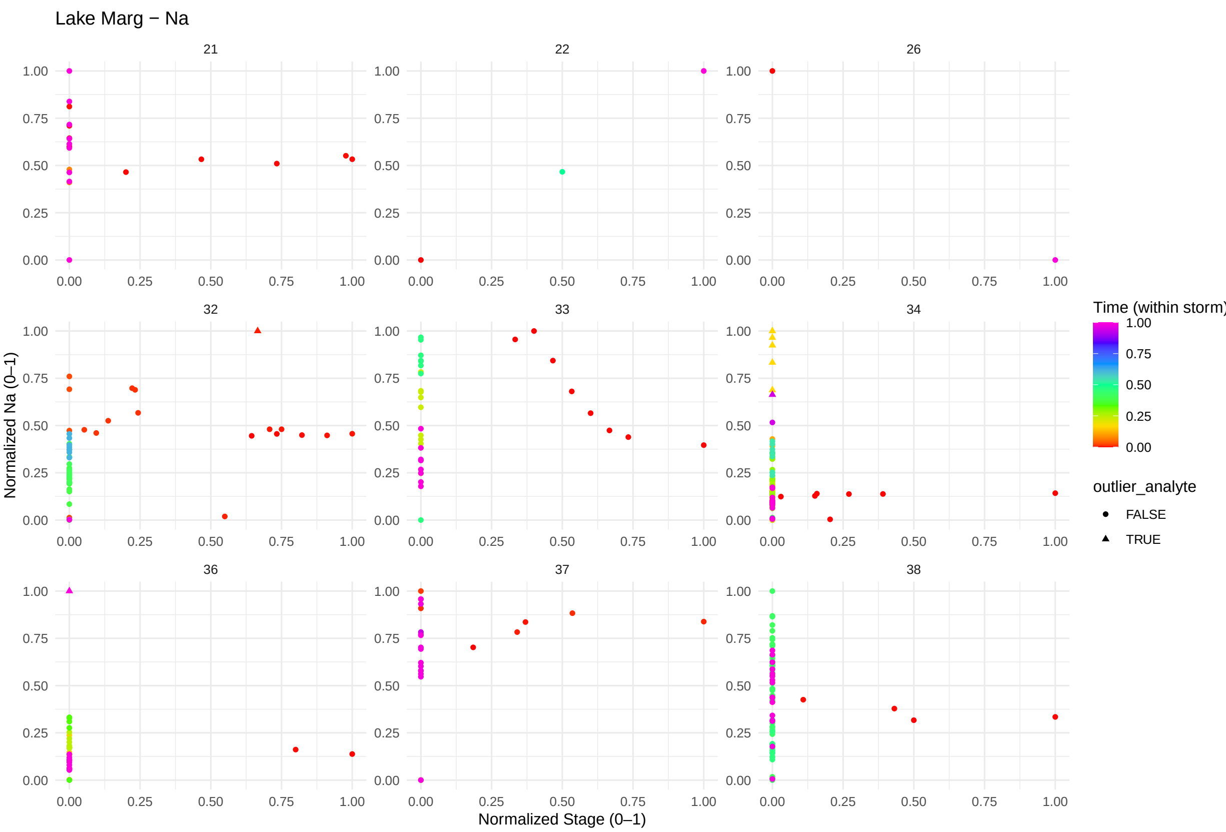

Lake Marg − DOC
21
22
26
32
33
34
36
37
38
Normalized DOC (0–1)
Normalized Stage (0–1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00
outlier_analyte
FALSE
TRUE


Lake Marg − NH4
21
22
26
32
33
34
Normalized NH4 (0–1)
Normalized Stage (0–1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00
outlier_analyte
FALSE
TRUE

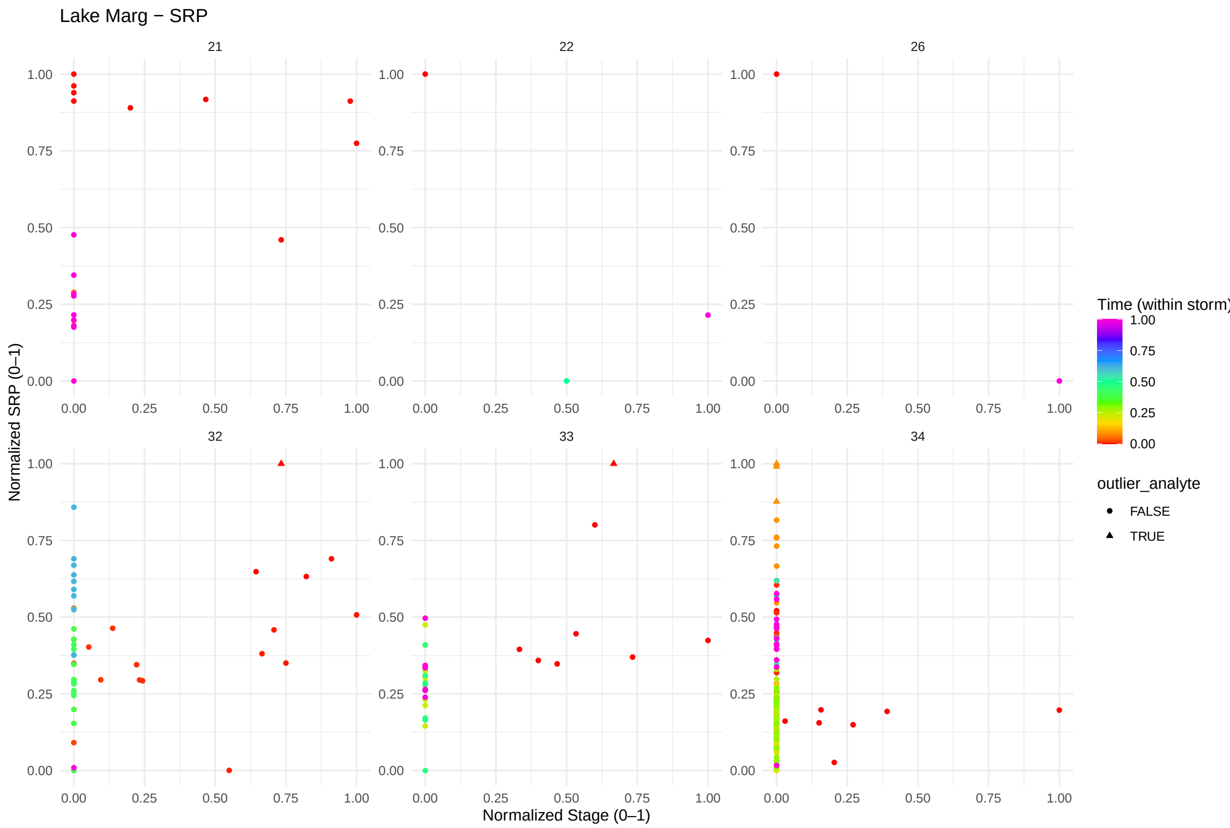

Lake Marg − NO2
21
22
26
32
33
34
Normalized NO2 (0–1)
Normalized Stage (0–1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00
outlier_analyte
FALSE
TRUE

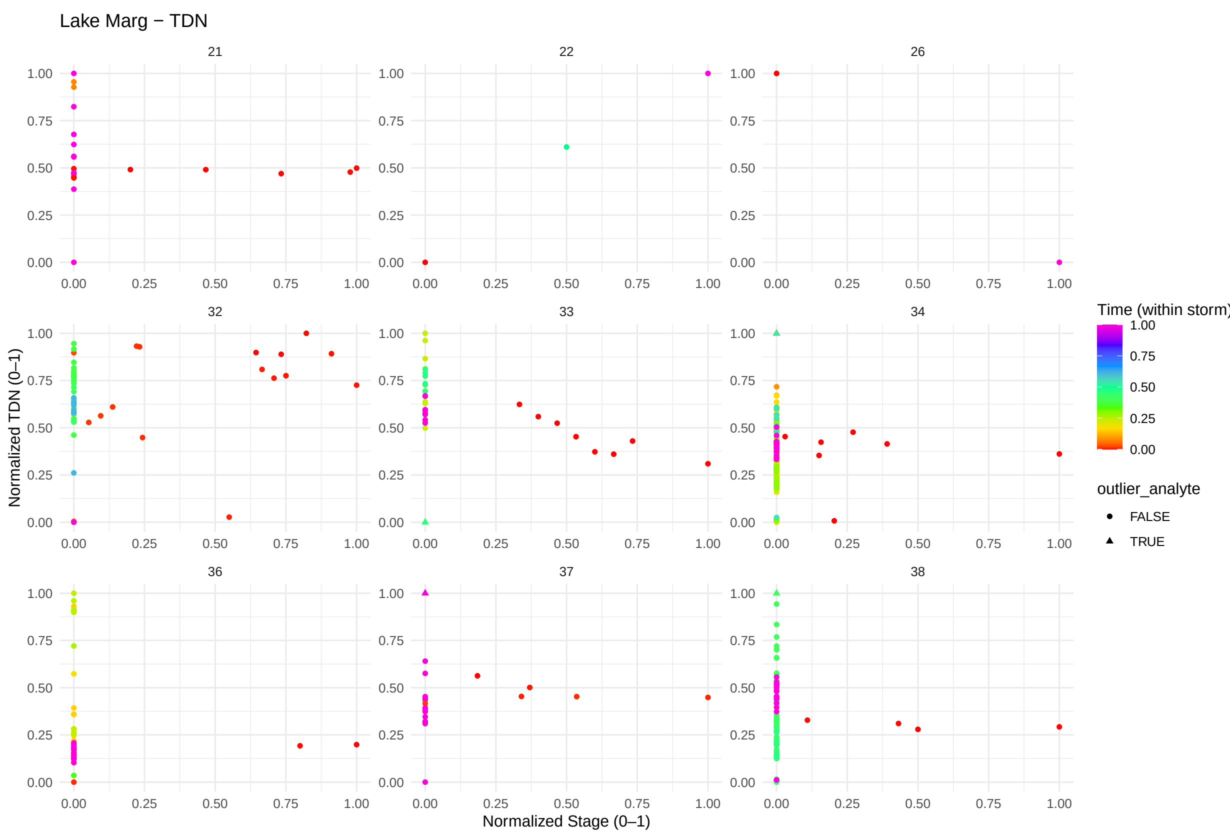


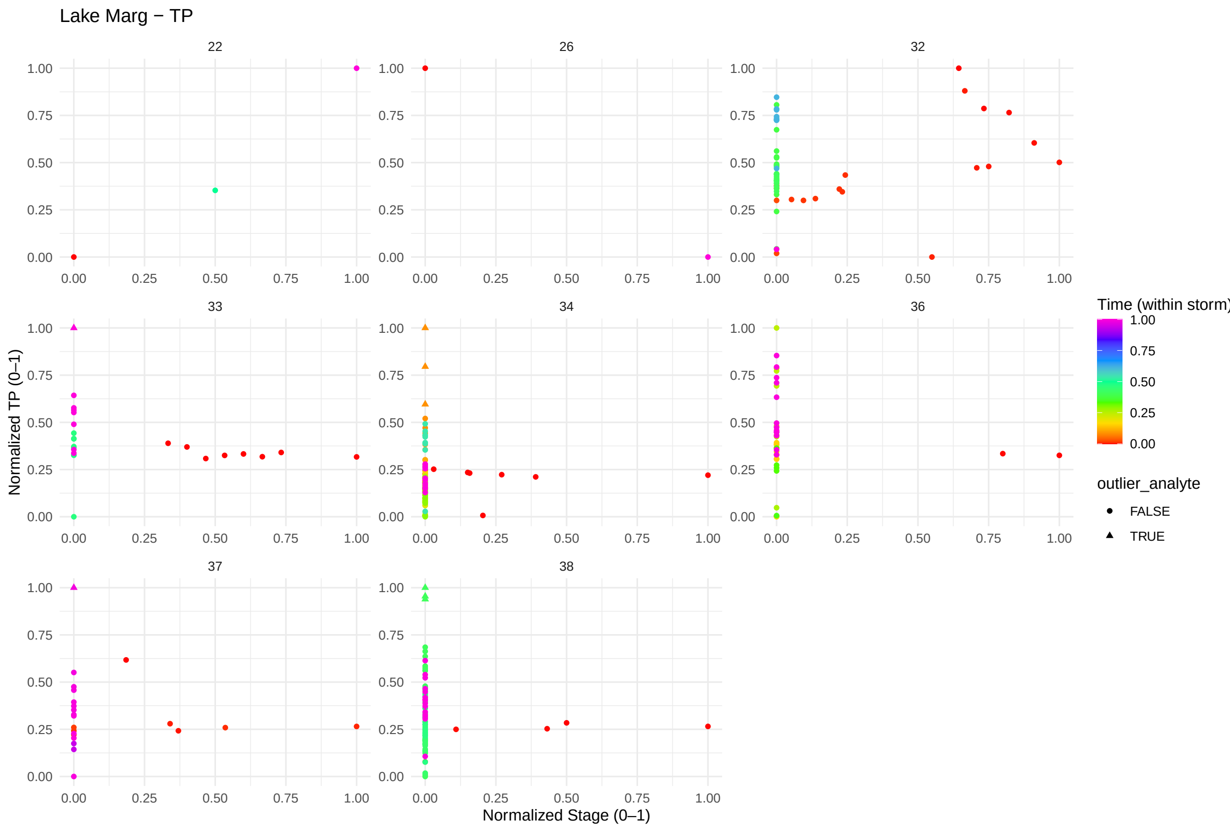
Lake Marg − NO3
21
22
26
32
33
34
Normalized NO3 (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE

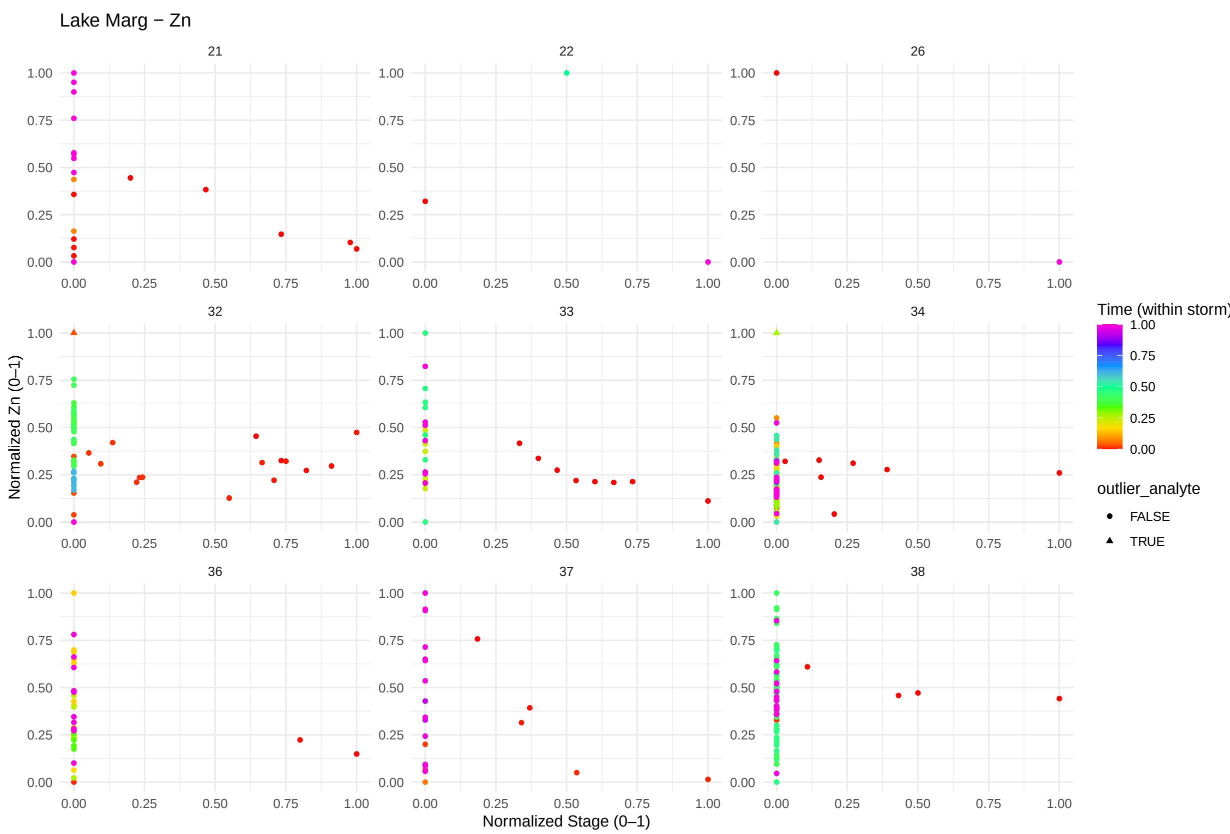


Lake Marg − Na
21
22
26
32
33
34
36
37
38
Normalized Na (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Lake Marg − SRP
21
22
26
32
33
34
Normalized SRP (0–1)
Normalized Stage (0–1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00
outlier_analyte
FALSE
TRUE


Lake Marg − TDN
21
22
26
32
33
34
36
37
38
Normalized TDN (0–1)
Normalized Stage (0–1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00
outlier_analyte
FALSE
TRUE

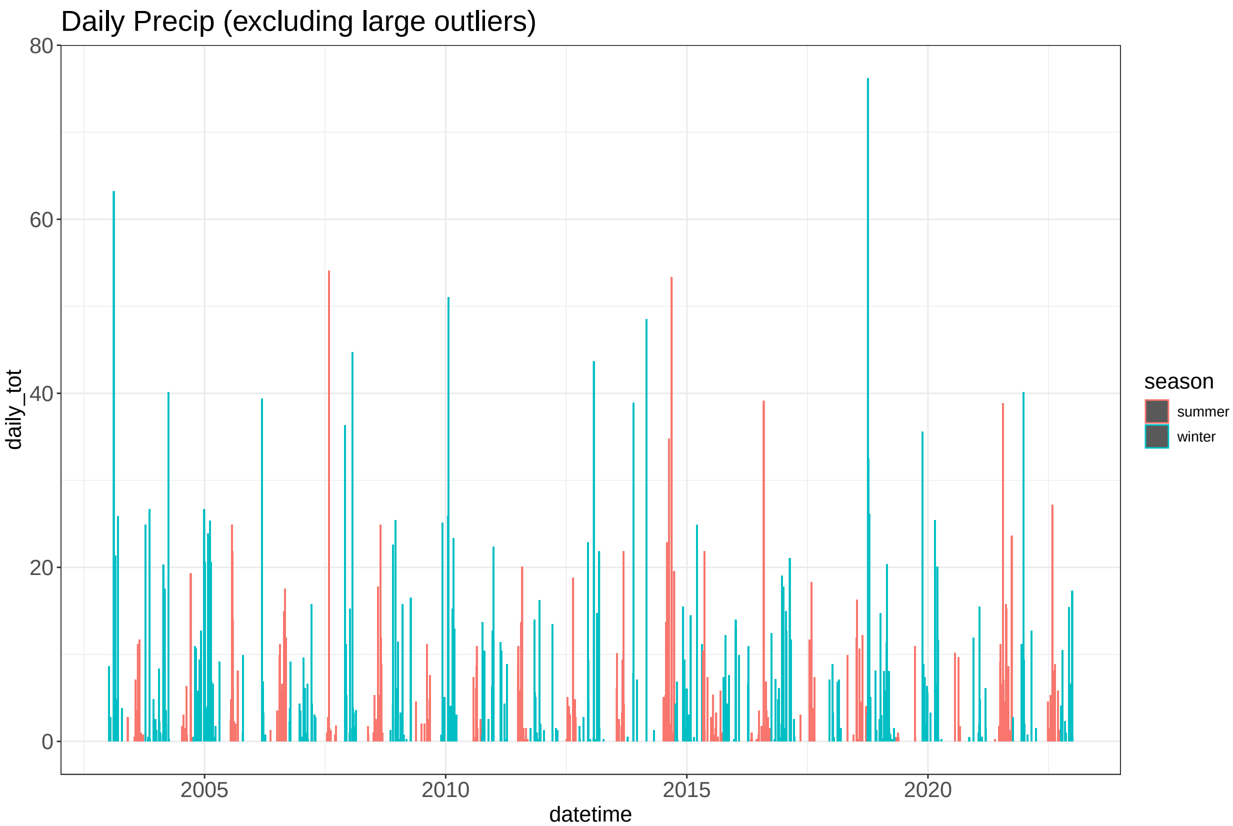

Lake Marg − TP
22
26
32
33
34
36
37
38
Normalized TP (0–1)
Normalized Stage (0–1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00
outlier_analyte
FALSE
TRUE

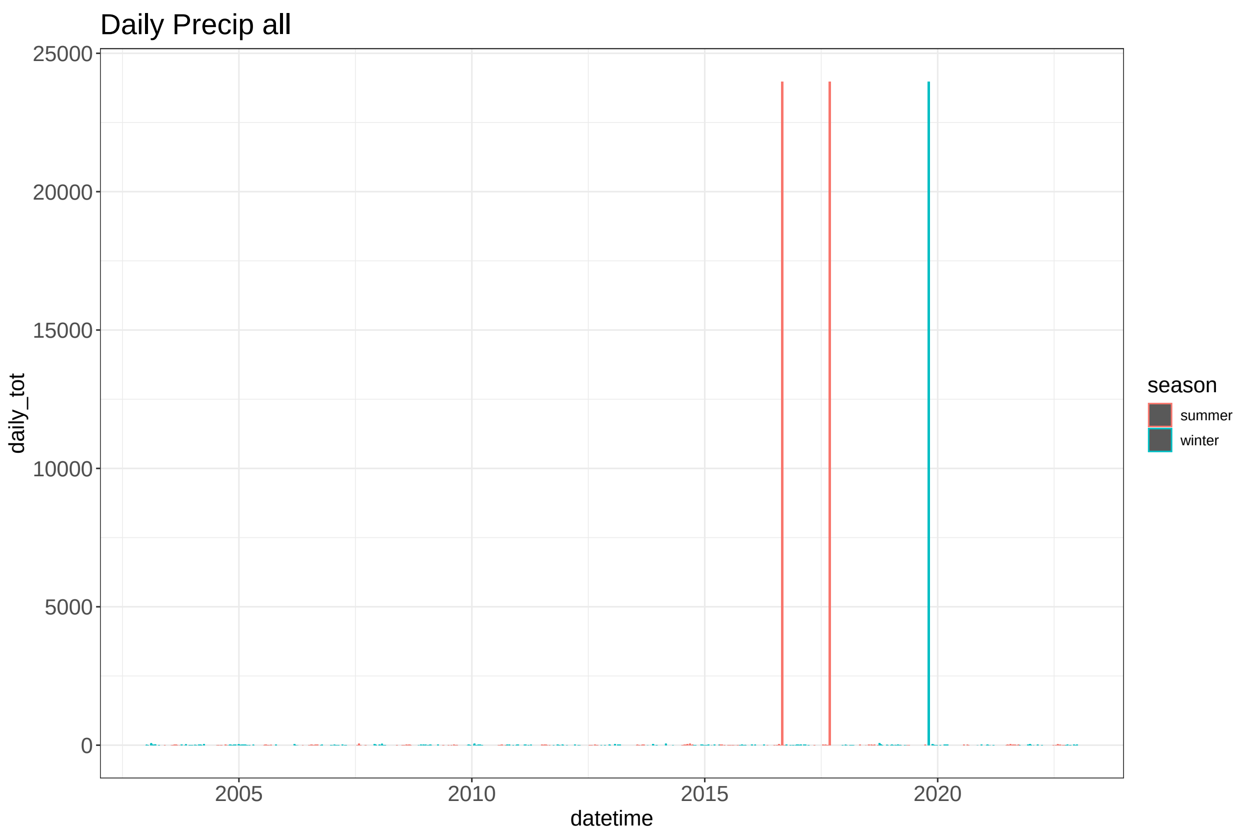

Lake Marg − Zn
21
22
26
32
33
34
36
37
38
Normalized Zn (0–1)
Normalized Stage (0–1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00
outlier_analyte
FALSE
TRUE

Precipitation

Precipitation data was collected from https://azmet.arizona.edu/azmet/27.htm - precipitation gage at the Desert Ridge station

Time series of precip

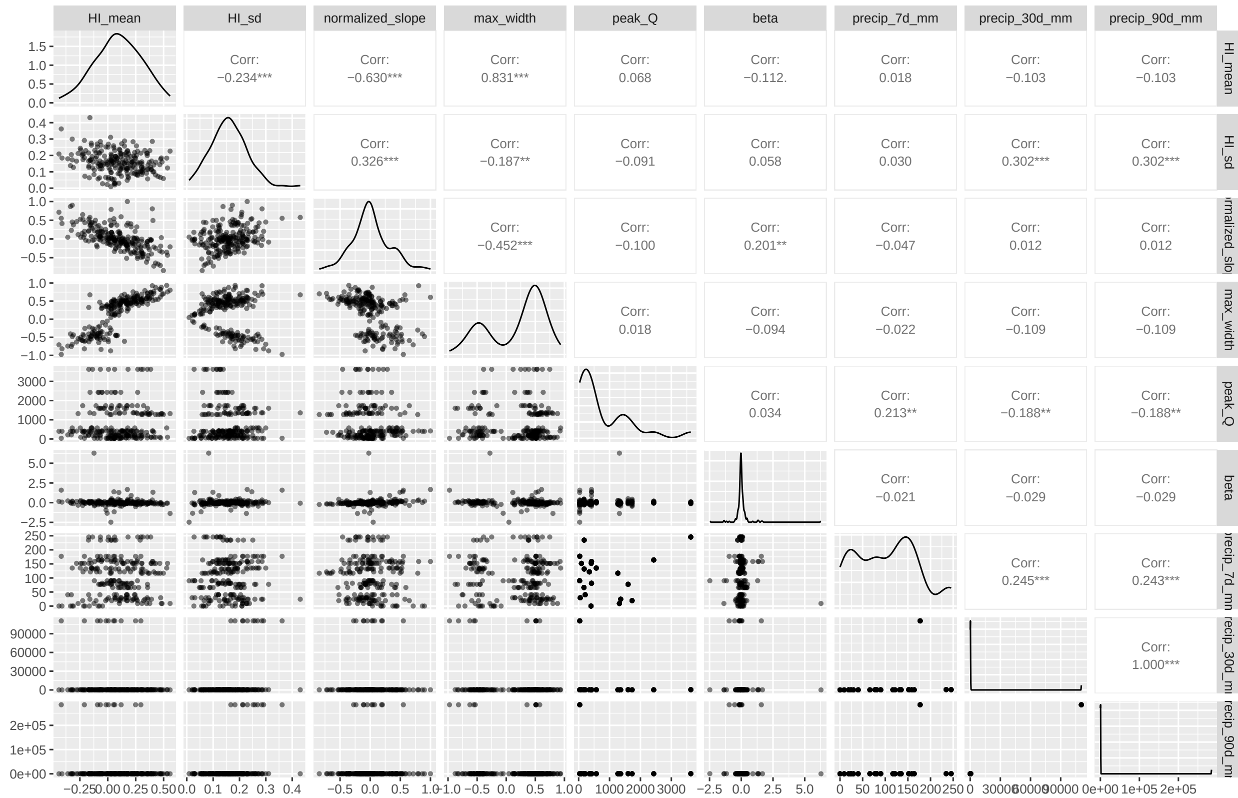




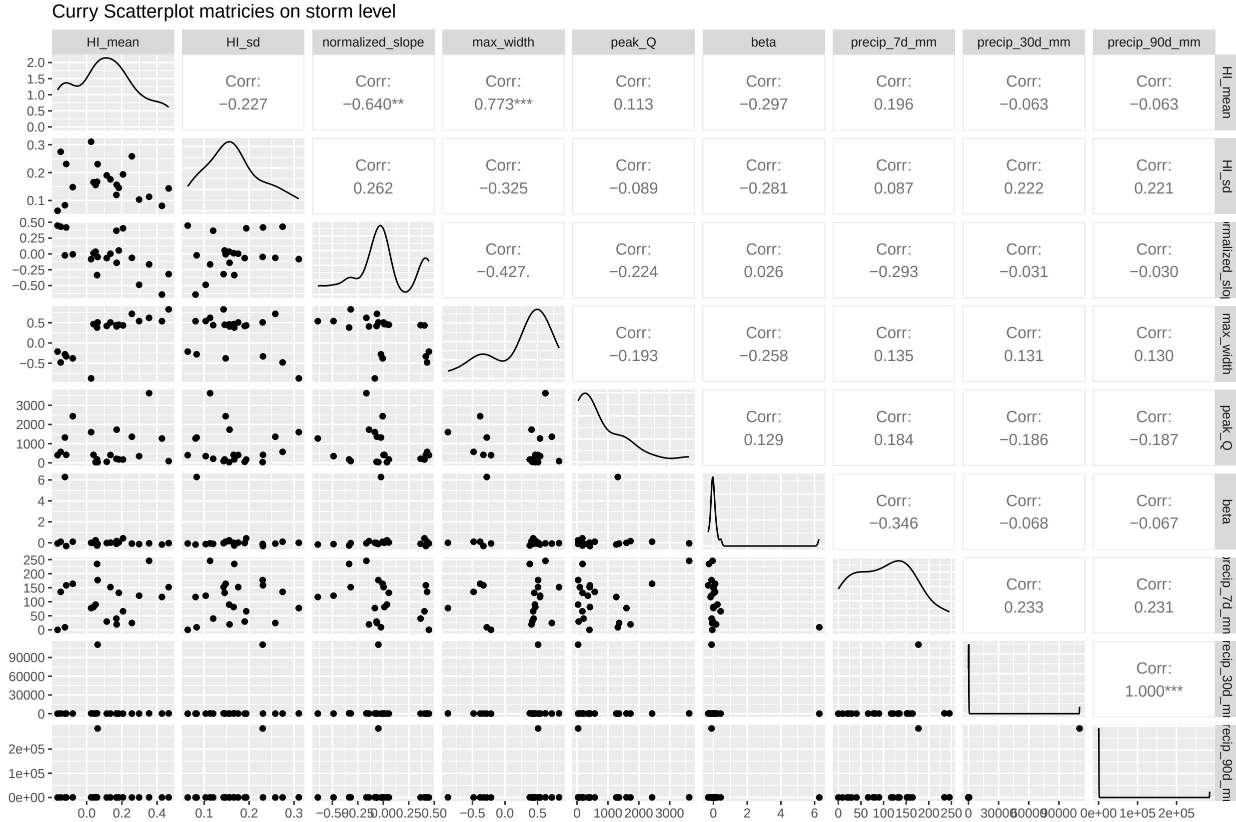
Scatterplot matricies

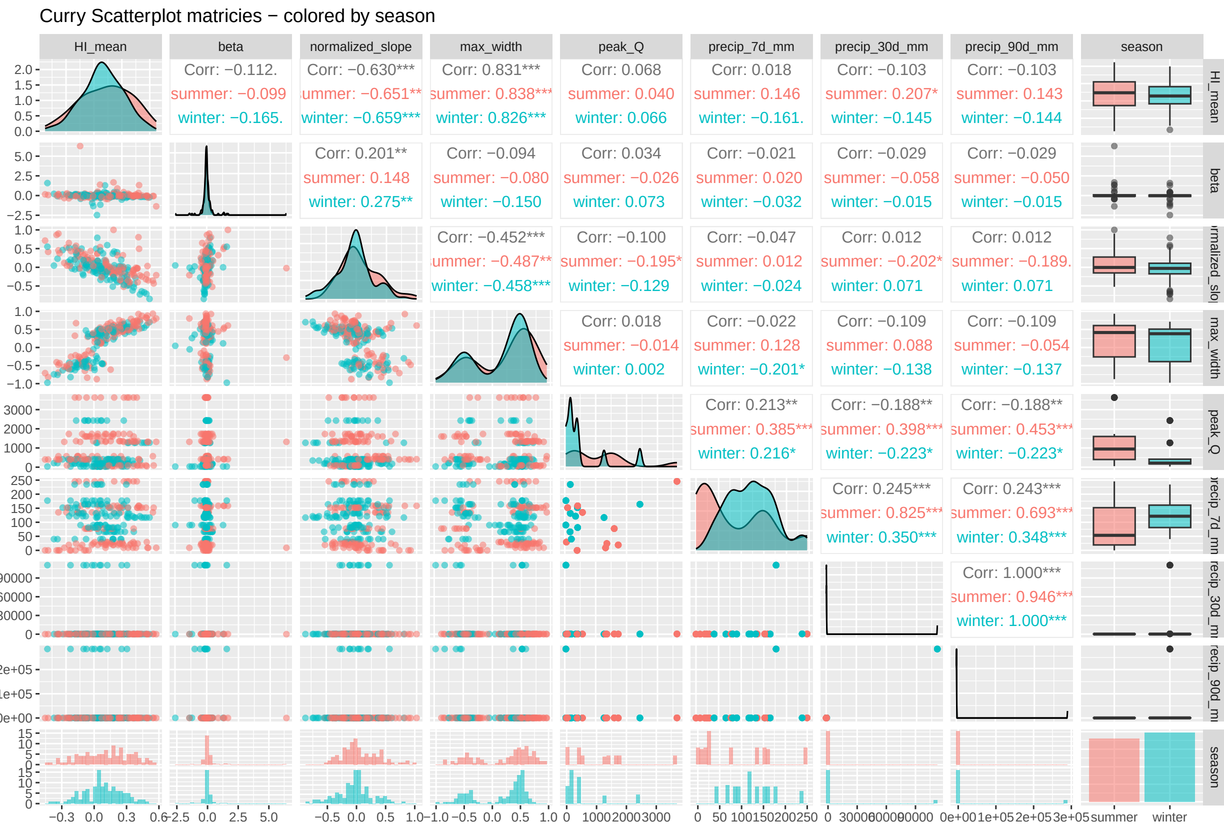
A few different options

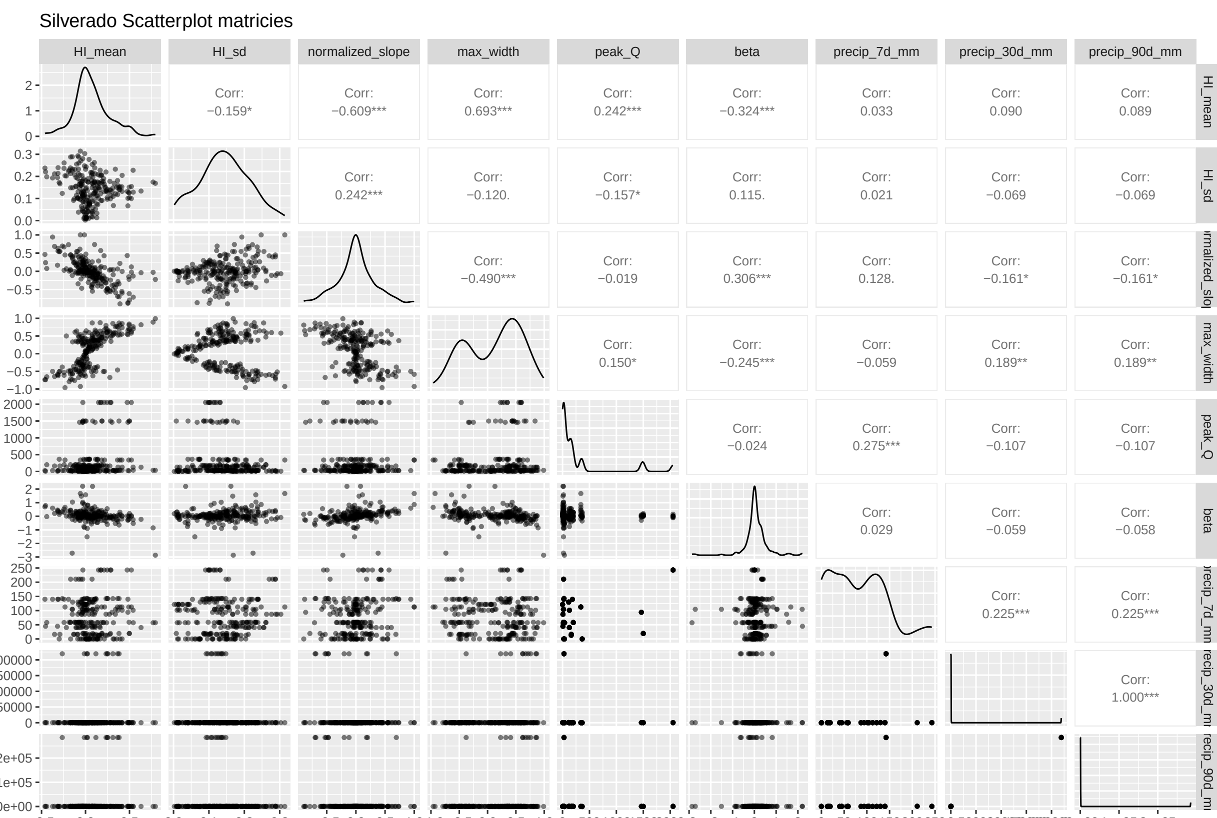
Curry Scatterplot matricies



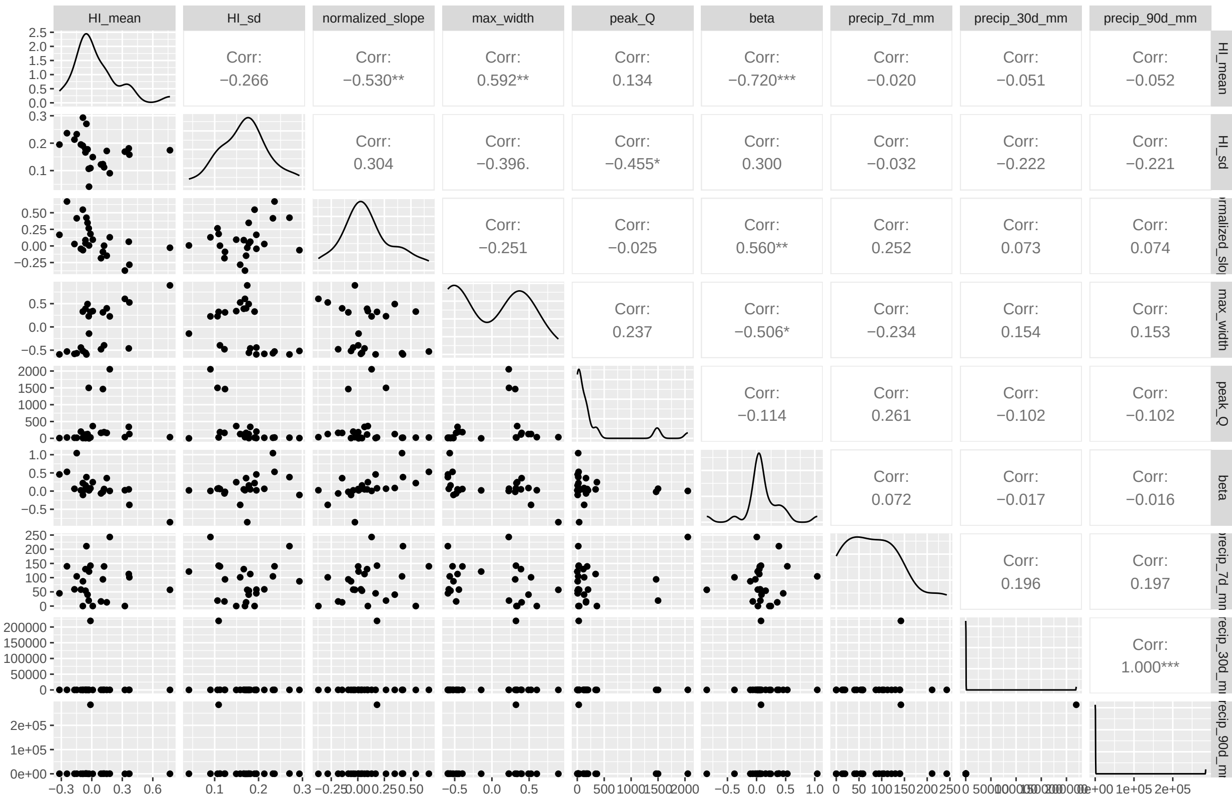
Curry Scatterplot matricies on storm level



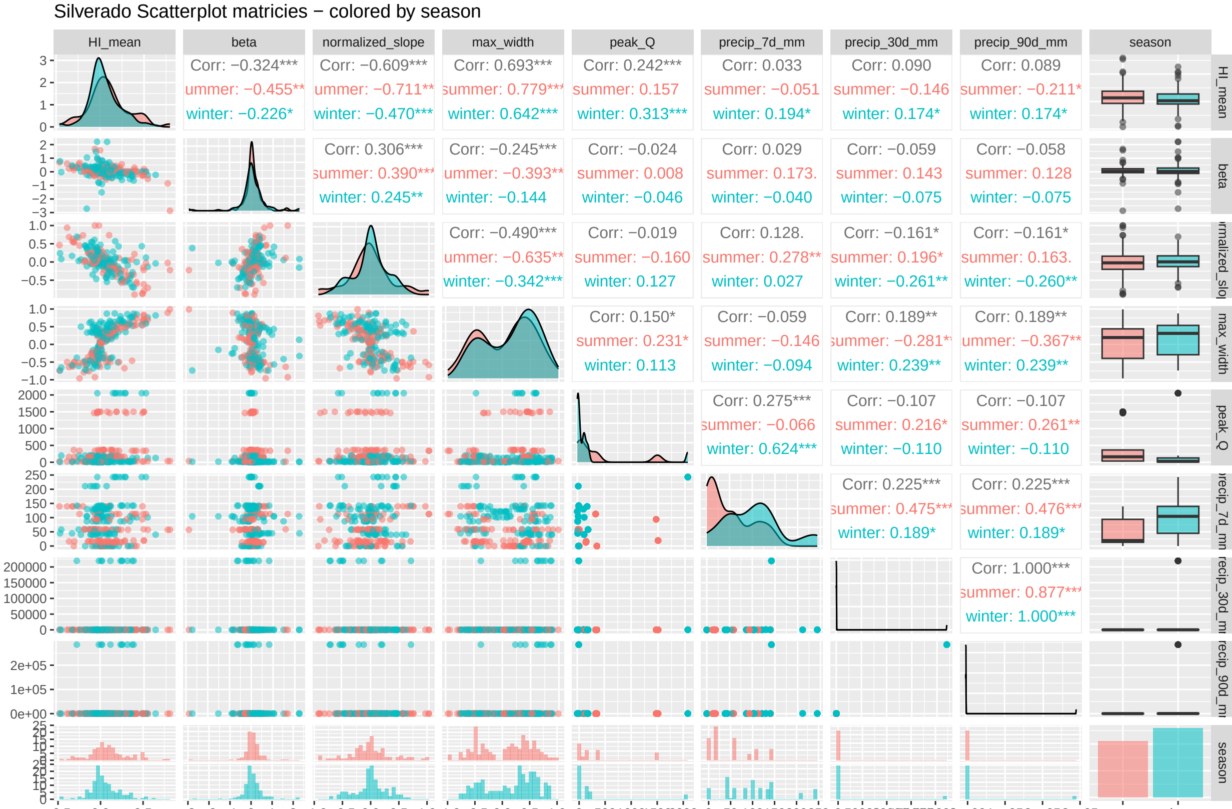

Curry Scatterplot matricies − colored by season
HI_mean
beta
normalized_slope
max_width
peak_Q
precip_7d_mm
precip_30d_mm
precip_90d_mm
season
Corr: −0.112.
summer: −0.099
winter: −0.165.
Corr: −0.630***
summer: −0.651***
winter: −0.659***
Corr: 0.831***
summer: 0.838***
winter: 0.826***
Corr: 0.068
summer: 0.040
winter: 0.066
Corr: 0.018
summer: 0.146
winter: −0.161.
Corr: −0.103
summer: 0.207*
winter: −0.145
Corr: −0.103
summer: 0.143
winter: −0.144
Corr: 0.201**
summer: 0.148
winter: 0.275**
Corr: −0.094
summer: −0.080
winter: −0.150
Corr: 0.034
summer: −0.026
winter: 0.073
Corr: −0.021
summer: 0.020
winter: −0.032
Corr: −0.029
summer: −0.058
winter: −0.015
Corr: −0.029
summer: −0.050
winter: −0.015
Corr: −0.452***
summer: −0.487***
winter: −0.458***
Corr: −0.100
summer: −0.195*
winter: −0.129
Corr: −0.047
summer: 0.012
winter: −0.024
Corr: 0.012
summer: −0.202*
winter: 0.071
Corr: 0.012
summer: −0.189.
winter: 0.071
Corr: 0.018
summer: −0.014
winter: 0.002
Corr: −0.022
summer: 0.128
winter: −0.201*
Corr: −0.109
summer: 0.088
winter: −0.138
Corr: −0.109
summer: −0.054
winter: −0.137
Corr: 0.213**
summer: 0.385***
winter: 0.216*
Corr: −0.188**
summer: 0.398***
winter: −0.223*
Corr: −0.188**
summer: 0.453***
winter: −0.223*
Corr: 0.245***
summer: 0.825***
winter: 0.350***
Corr: 0.243***
summer: 0.693***
winter: 0.348***
Corr: 1.000***
summer: 0.946***
winter: 1.000***
summer
winter


Silverado Scatterplot matricies
HI_mean
HI_sd
normalized_slope
max_width
peak_Q
beta
precip_7d_mm
precip_30d_mm
precip_90d_mm
Corr: −0.159*
Corr: −0.609***
Corr: 0.693***
Corr: 0.242***
Corr: −0.324***
Corr: 0.033
Corr: 0.090
Corr: 0.089
Corr: 0.242***
Corr: −0.120.
Corr: −0.157*
Corr: 0.115.
Corr: 0.021
Corr: −0.069
Corr: −0.069
Corr: −0.490***
Corr: −0.019
Corr: 0.306***
Corr: 0.128.
Corr: −0.161*
Corr: −0.161*
Corr: 0.150*
Corr: −0.245***
Corr: −0.059
Corr: 0.189**
Corr: 0.189**
Corr: −0.024
Corr: 0.275***
Corr: −0.107
Corr: −0.107
Corr: 0.029
Corr: −0.059
Corr: −0.058
Corr: 0.225***
Corr: 0.225***
Corr: 1.000***

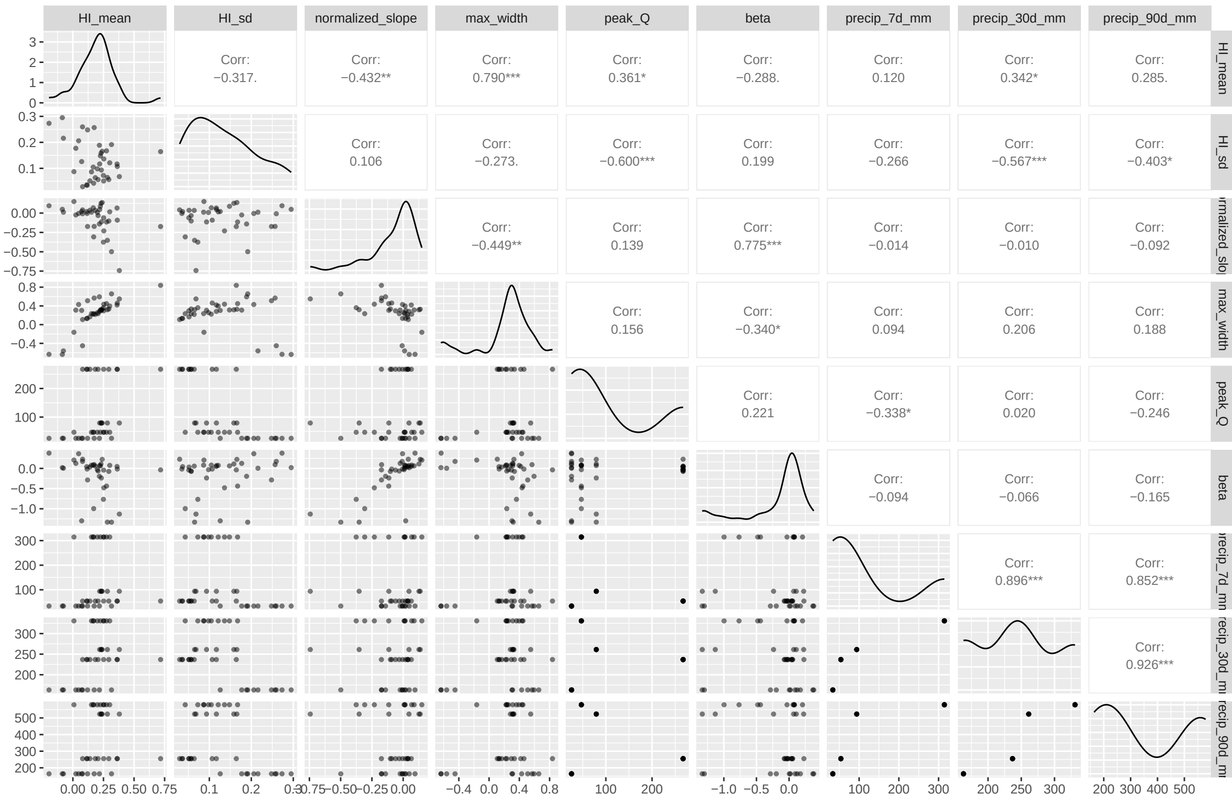
Silverado Scatterplot matricies on storm level



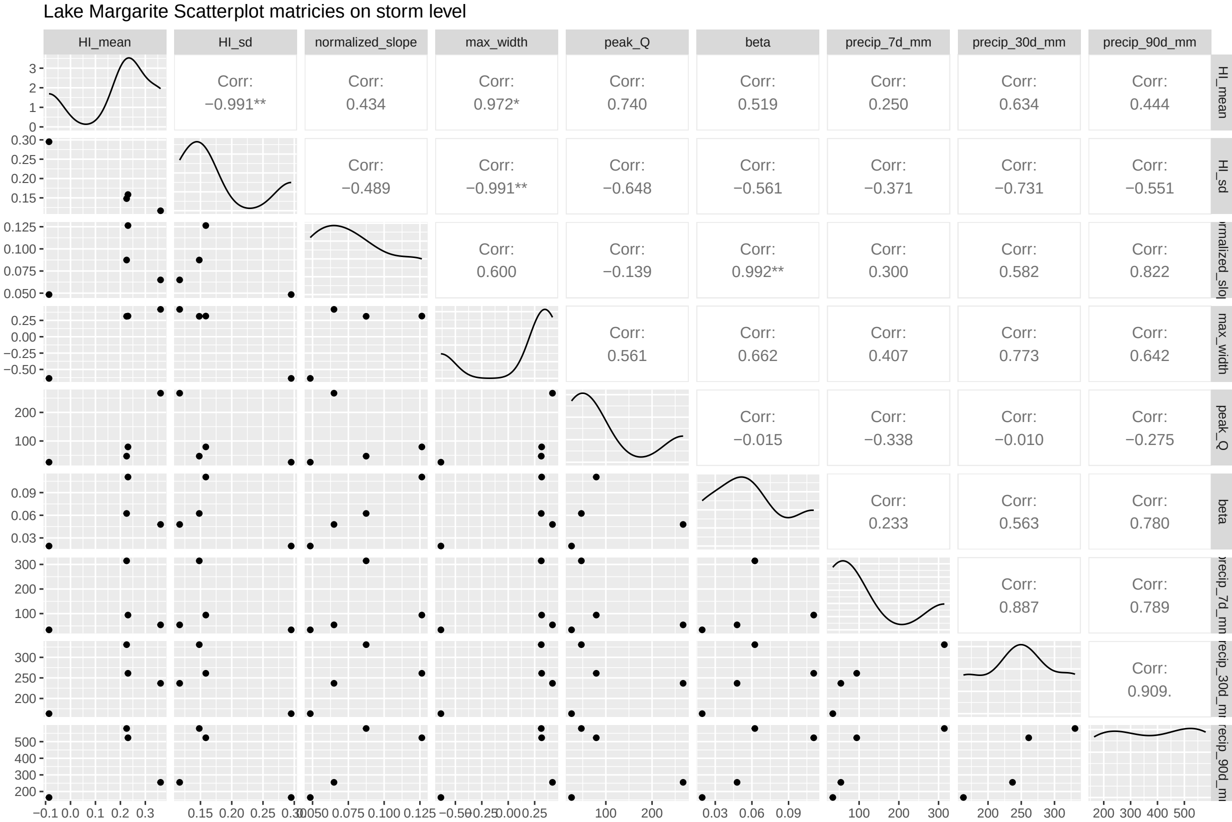
Silverado Scatterplot matricies − colored by season

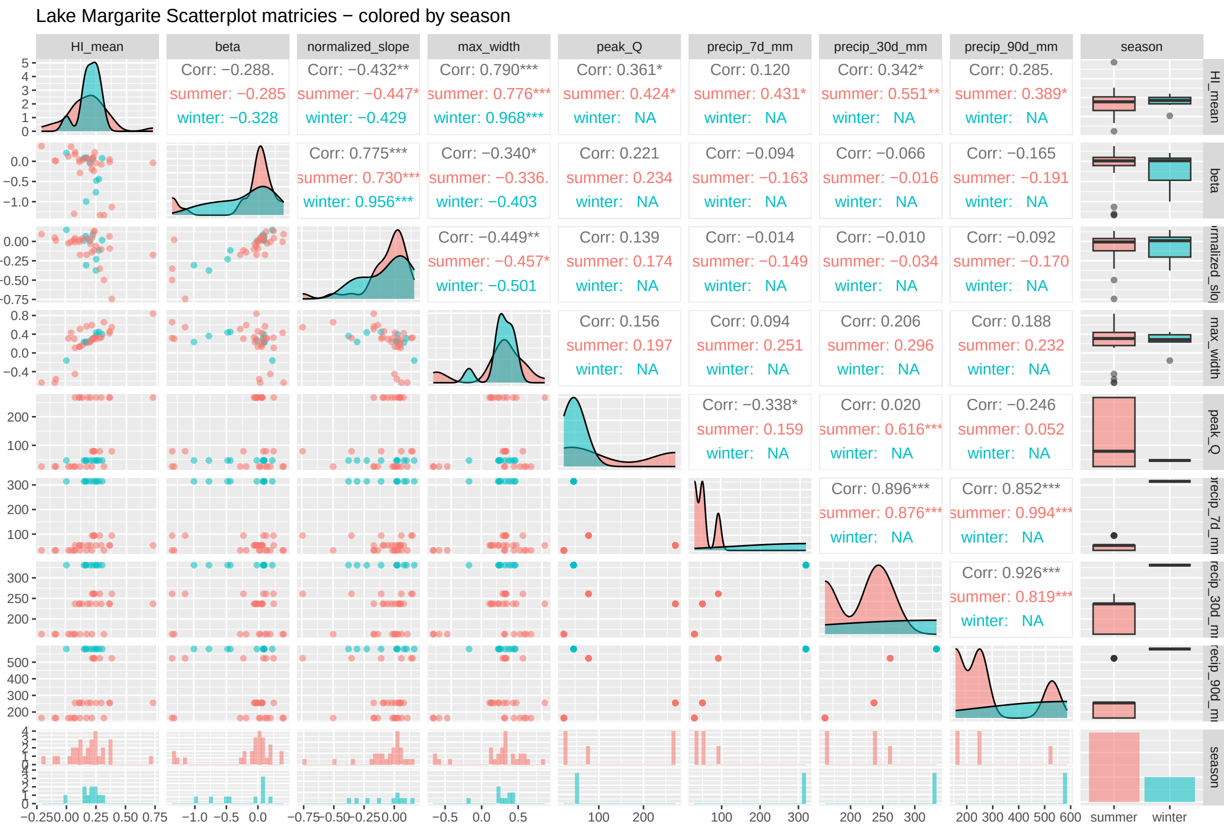


Lake Margarite Scatterplot matricies

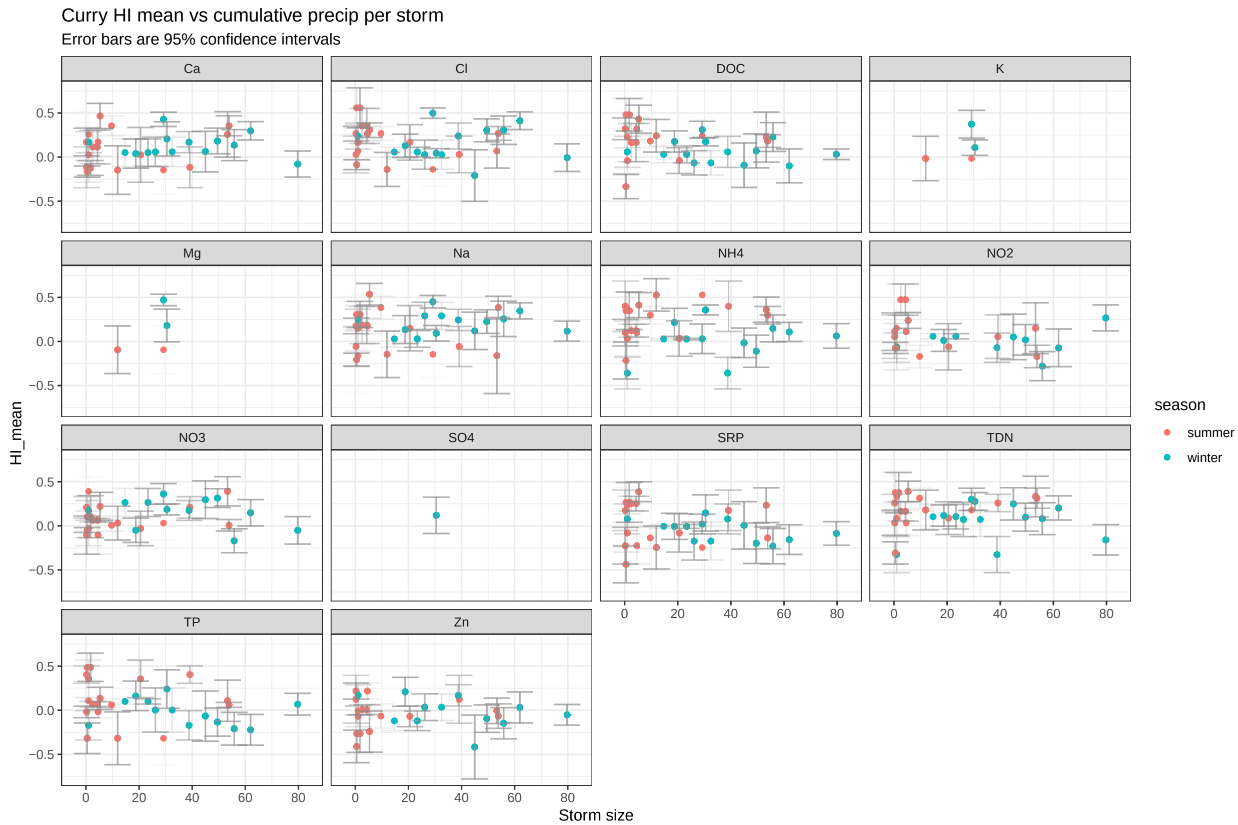


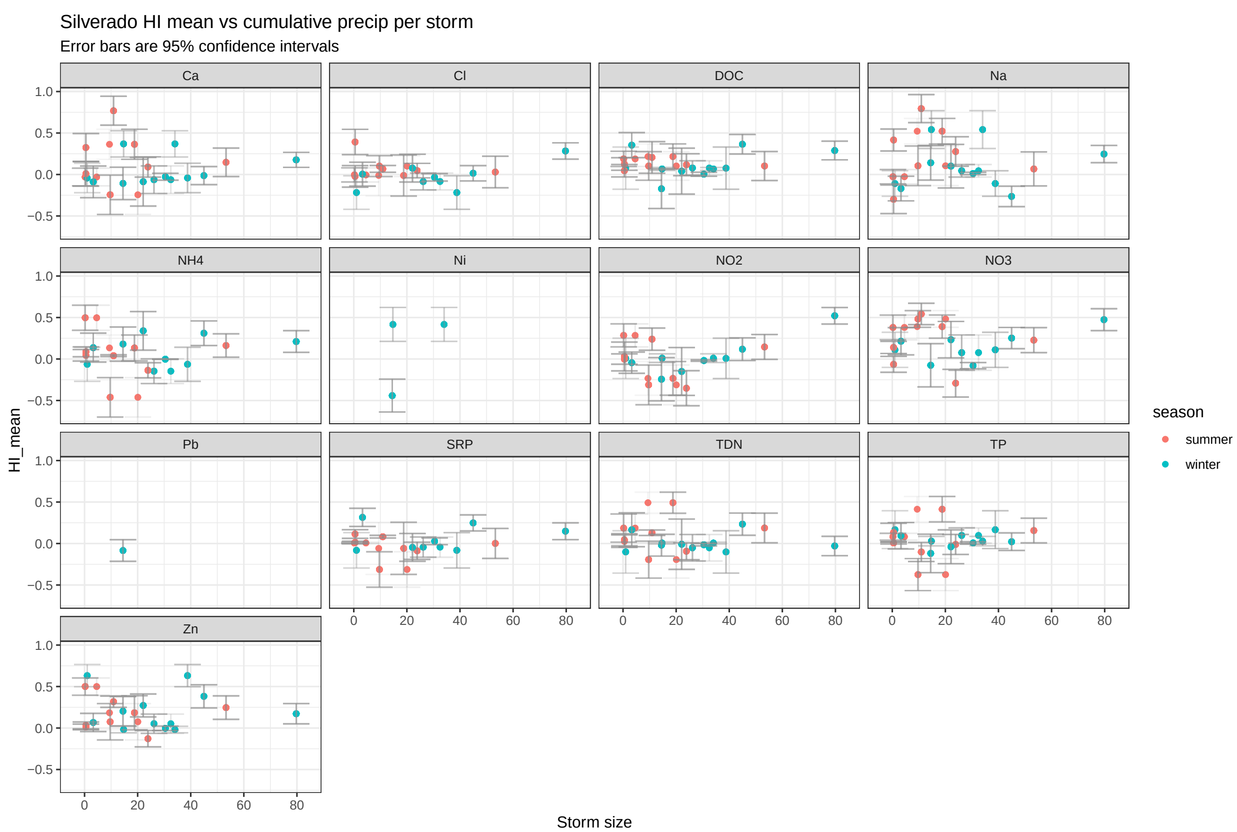
Lake Margarite Scatterplot matricies on storm level

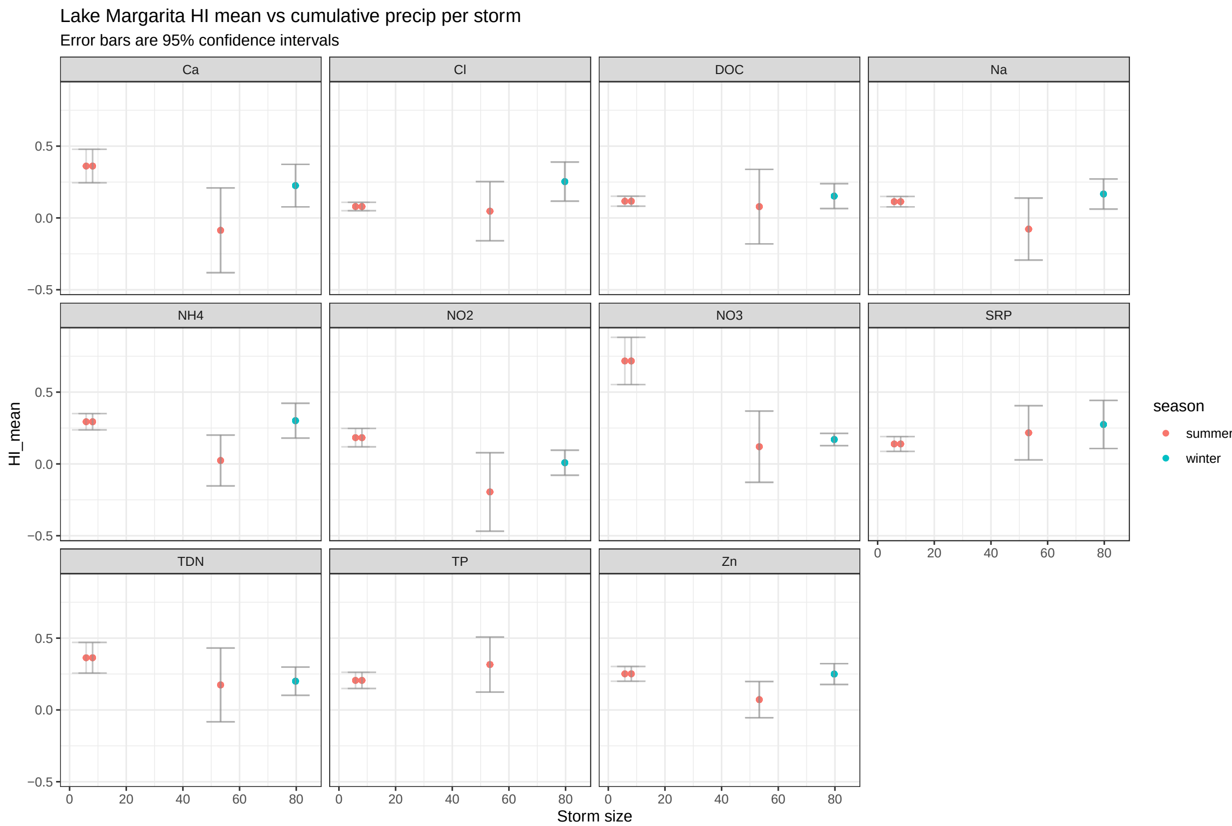



Lake Margarite Scatterplot matricies − colored by season
HI_mean
beta
normalized_slope
max_width
peak_Q
precip_7d_mm
precip_30d_mm
precip_90d_mm
season
Corr: −0.288.
summer: −0.285
winter: −0.328
Corr: −0.432**
summer: −0.447*
winter: −0.429
Corr: 0.790***
summer: 0.776***
winter: 0.968***
Corr: 0.361*
summer: 0.424*
winter: NA
Corr: 0.120
summer: 0.431*
winter: NA
Corr: 0.342*
summer: 0.551**
winter: NA
Corr: 0.285.
summer: 0.389*
winter: NA
Corr: 0.775***
summer: 0.730***
winter: 0.956***
Corr: −0.340*
summer: −0.336.
winter: −0.403
Corr: 0.221
summer: 0.234
winter: NA
Corr: −0.094
summer: −0.163
winter: NA
Corr: −0.066
summer: −0.016
winter: NA
Corr: −0.165
summer: −0.191
winter: NA
Corr: −0.449**
summer: −0.457*
winter: −0.501
Corr: 0.139
summer: 0.174
winter: NA
Corr: −0.014
summer: −0.149
winter: NA
Corr: −0.010
summer: −0.034
winter: NA
Corr: −0.092
summer: −0.170
winter: NA
Corr: 0.156
summer: 0.197
winter: NA
Corr: 0.094
summer: 0.251
winter: NA
Corr: 0.206
summer: 0.296
winter: NA
Corr: 0.188
summer: 0.232
winter: NA
Corr: −0.338*
summer: 0.159
winter: NA
Corr: 0.020
summer: 0.616***
winter: NA
Corr: −0.246
summer: 0.052
winter: NA
Corr: 0.896***
summer: 0.876***
winter: NA
Corr: 0.852***
summer: 0.994***
winter: NA
Corr: 0.926***
summer: 0.819***
winter: NA
summer
winter

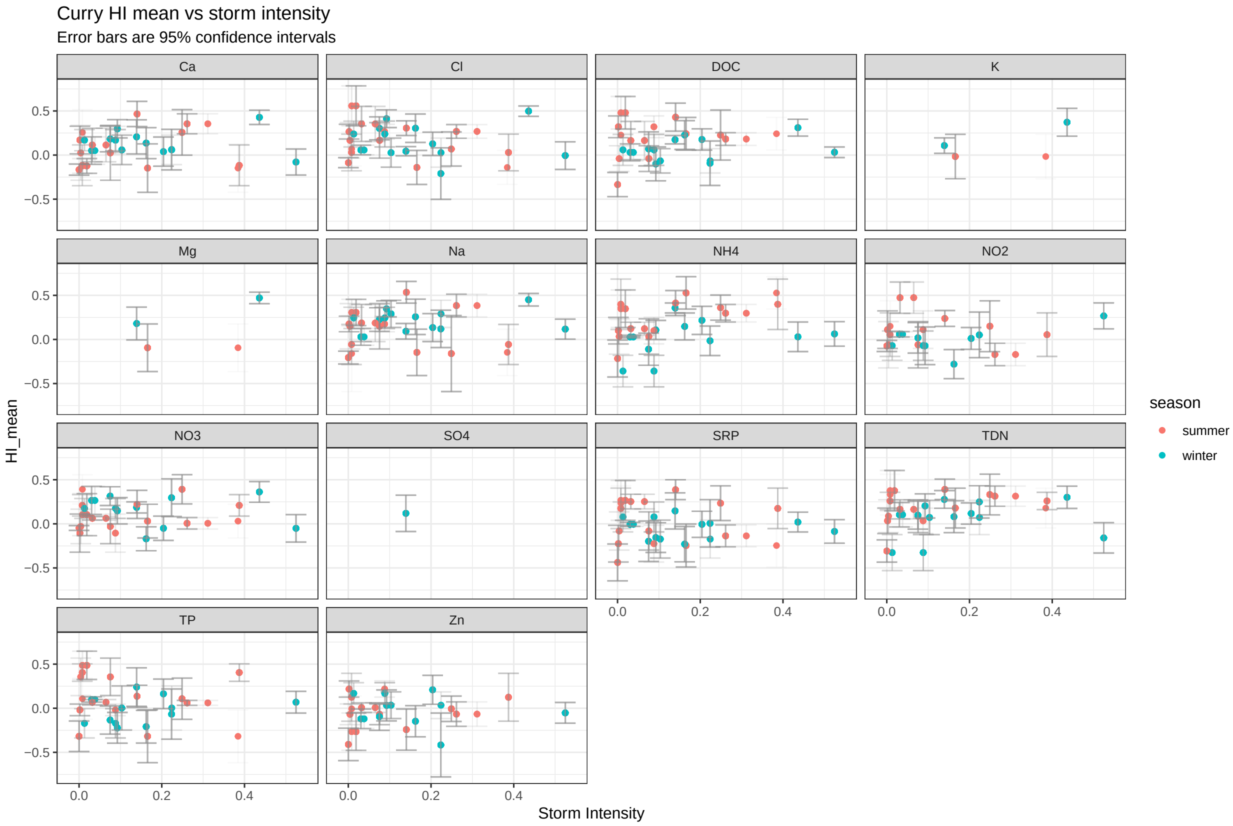
Storm size = cumulative precipitation (mm) per storm

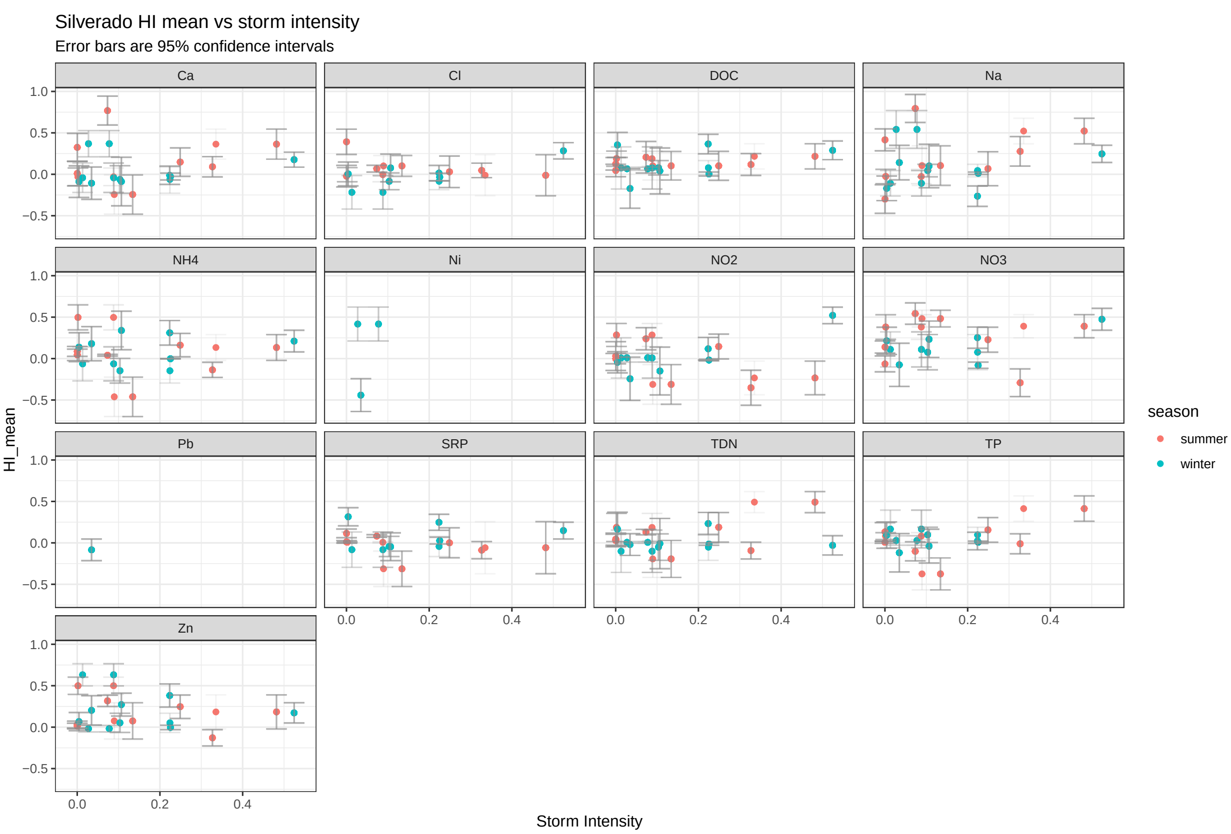


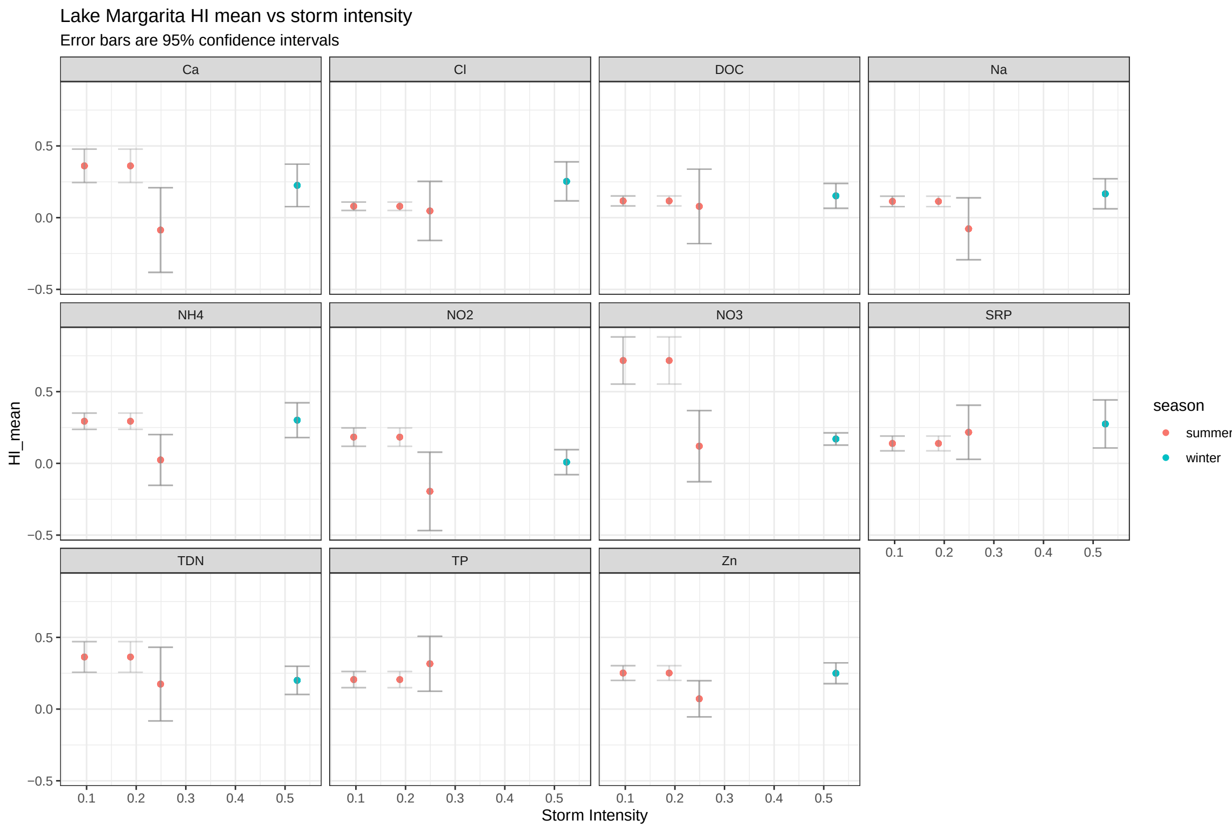



Lake Margarita HI mean vs cumulative precip per storm
Error bars are 95% confidence intervals
Ca
Cl
DOC
Na
NH4
NO2
NO3
SRP
TDN
TP
Zn
HI_mean
Storm size
season
summer
winter
0.5
0.0
−0.5
0
20
40
60
80

Storm intensity = cumulative precipitation/storm duration







Lake Margarita HI mean vs storm intensity
Error bars are 95% confidence intervals
Ca
Cl
DOC
Na
NH4
NO2
NO3
SRP
TDN
TP
Zn
HI_mean
Storm Intensity
season
summer
winter
0.5
0.0
−0.5
0.1
0.2
0.3
0.4
0.5